

Machine learning methods for continuous glucose monitoring data and their potential in cardiovascular risk assessment

PhD dissertation

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Disclosures and Notes

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Notes on website version

This dissertation is available for viewing online at HeleneThomsen.github.io/dissertation/ or by scanning the QR code in the lower right-hand corner of the title page.

Acknowledgements

Preface

Outline of the dissertation

Papers in the dissertation

Study I

Thomsen, H. B., Lebiecka-Johansen, B., Nørgaard, O., Andersen, T. H., Andersen, S. T., Fagherazzi, G., Hulman, A., & Isaksen, A. A. (2026). Continuous Glucose Monitoring–Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review. *JMIR Diabetes*, 11(1), e89374. <https://doi.org/10.2196/89374>

Study II

Thomsen, H. B., Andersen, S. T., Fagherazzi, G., Isaksen, A. A. & Hulman, A. (2026). Foundation model–derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk. (Under review at *JMIR AI*, (7.30.2026), pending for preprint at *JMIR preprints*)

Study III

Thomsen, H. B., Li, L. Y., Isaksen, A. A., Lebiecka-Johansen, B., Bour, C., Fagherazzi, G., Doorn, W. P. T. M. van, Varga, T. V., & Hulman, A. (2025). Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes. *PLOS Digital Health*, 4(6), e0000918. <https://doi.org/10.1371/journal.pdig.0000918>

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Abbreviations

1 Introduction

1.1 Diabetes

- Check the sources from ADA_2026_ch2 to see if you can elaborate on the race thing and cite the original articles and not the summary ADA gives

Diabetes is a metabolic disease characterized by high blood glucose levels known as hyperglycemia.¹ The disease is one of the fastest growing global health challenges.² In 2021 it was estimated that 537 million people lived with diabetes, and it is expected to reach 643 million by 2030.² The most common forms of chronic diabetes are type 1 diabetes (T1D) and type 2 diabetes (T2D).¹ T1D is an autoimmune disease characterized by the destruction of the pancreatic islet β -cells, leading to an absolute insulin deficiency.³ Approximately 85%-95% of all people with diabetes in developing countries have T2D.¹ It is the most common form of diabetes and in contrast to T1D it generally begins with insulin resistance in which the body's cells become less responsive to insulin.^{4, Sachs2023?} Initially, the pancreas compensates by increasing insulin secretion, but over time progressive β -cell dysfunction results in inadequate insulin production to meet the body's demands.^{1,4,5} T2D is the most which is is Despite their distinct underlying pathophysiology, both T1D and T2D are characterized by chronic hyperglycemia.

Diabetes can be diagnosed based on a bloodtest to check hemoglobin A1c (HbA1c) levels or through plasma glucose criteria (Table 1.1).^{3, ADA2023_ch6?}

Although both plasma glucose tests and HbA1c tests are appropriate for diabetes screening, the HbA1c test has several advantages.³ Some of the benefits are that it requires no fasting and it is less sensitive to stress and changes in nutrition.³ HbA1c is an indirect measure of blood glucose levels because glucose binds hemoglobin over ~ 120 days lifespan of erythrocytes (red blood cells).³ However HbA1c testing is not always accessible, affordable or suitable (e.g., for people with anemia or under HIV-treatment).³ With the guidelines recommendation of using HbA1c, a debate has sparked on the controversy of using a set thresholds as it is well known that racial differences in HbA1c exists.⁶ Black individuals have higher HbA1c levels than Hispanic and non-Hispanic White people with diabetes,^{3,7} however studies show that the association between HbA1c and risk of complications appear to be similar between ethnic race groups.³

1.2 Blood Glucose Monitoring

HbA1c values are also a very important part of diabetes management. The treatment goal is for HbA1c to be below 53 mmol/mol for non-pregnant adults without any hyperglycemia or hypoglycemia (low blood glucose levels). However HbA1c goals should be individualized to fit each individuals specific situation HbA1c can vary from e.g., <48 mmol/mol to none.^{8,9} There are two tools available for self-assessment of glycemia, the self-monitoring blood glucose (SMBG) device measuring capillary blood with finger-sticks, and the continuous glucose monitoring device (CGM) measuring interstitial fluid continuously.^{9,10} Both monitoring devices have shown to

Table 1.1: Diagnostic tests for diabetes

Test	Criteria	Description
Fasting plasma glucose	≥ 7.0 [mmol/L]	Fasting is defined as no calorie intake for at least eight hours.
2-hour plasma glucose	≥ 11.1 [mmol/L]	Glucose is measured 2 hours after oral ingestion of 75g glucose dissolved in water (OGTT)
Random plasma glucose	≥ 11.1 [mmol/L]	Individuals with classic symptoms of hyperglycemia (e.g., dehydration and unintentional weight loss) or hyperglycemic crisis (e.g., diabetic ketoacidosis) together with a plasma test taken any time of day.
Glycated hemoglobin (HbA1c)	≥ 48 [mmol/mol]	

improve glycemic control in people with diabetes, however the CGM is superior when it comes to immediate information and alerts on asymptomatic hypo- and hyperglycemic events and provide a comprehensive glucose profile including both pre- and postprandial blood glucose measurements.^{9,11,12}

1.3 Continuous Glucose Monitoring Devices

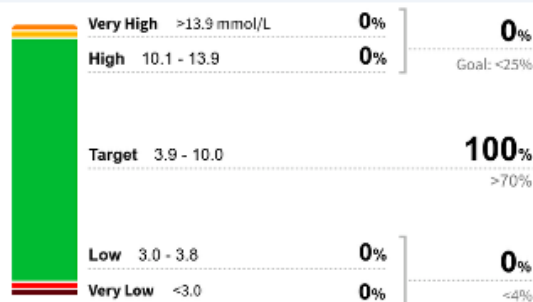
The CGM devices measures the glucose in the interstitial fluid, which is the fluid around and between cells. It corresponds with plasma glucose although a lag can appear when glucose levels rise or fall rapidly.^{ADA2026_ch7?} Use of CGM devices has shown to improve glycemic control in both T1D and T2D with improvements in HbA1c levels and more time spent in target range (3.9-10 mmol/L) regardless of age, sex, education and income.^{ADA2026_ch6?,ADA2026_ch7?} CGM devices focuses on monitoring and it is possible to get an ambulatory glucose profile (AGP) report from you device (Figure 1.1). The AGP report shows summary statistics over all your data (usually a two weeks period). Some of the most important management measures from the CGM is time in range (TIR), and it is the time spent within normal range in percentage over a given time (often two weeks). Other CGM-derived metrics are time above range (TAR), time below range (TBR), glucose management indicator (GMI), coefficient of variation (CV) and the mean glucose (MBG), number of days CGM was worn and the percentage of time the CGM was active (Table 1.2).¹¹

Glucose Pattern Insights

Selected Dates: 11 Apr 2024 - 24 Apr 2024 (14 Days)

LibreView

Time in Ranges



Glucose Statistics

Average Glucose

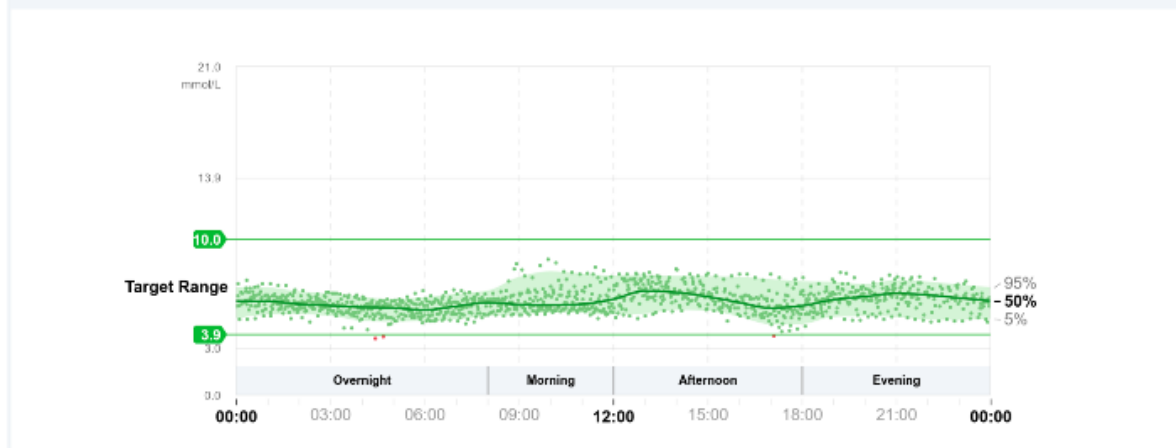
6.0 mmol/L Goal: ≤8.6 mmol/L

Glucose Management Indicator (GMI)

Approximate A1C level based on average CGM glucose level.

5.9% Goal: ≤7.0% | **41** mmol/mol Goal: ≤53 mmol/mol

Glucose Patterns (14 Days)



DAILY GLUCOSE PROFILES

Each daily profile represents a midnight to midnight period with the date displayed in the upper left corner.

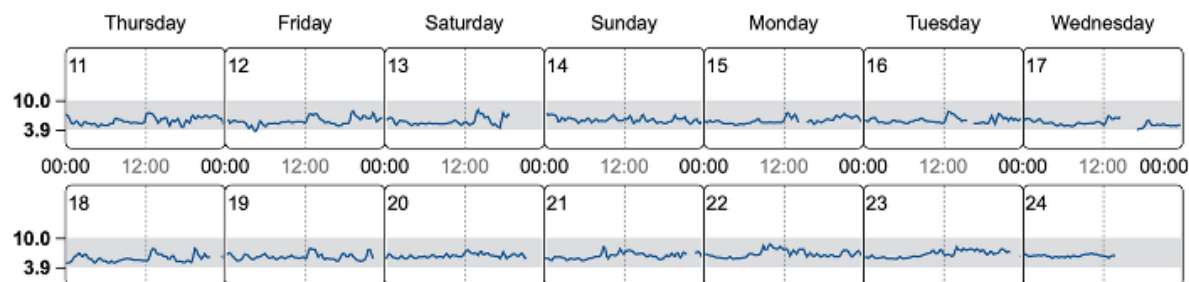


Figure 1.1: Ambulatory glucose profile report from female without diabetes. Data points are from the author herself using an expired intermittently-scanned FreeStyle LibreLink sensor (Manuel data upload).

1.4 Management

1.5 Glycemic markers and glucose monitoring

- ☒ HbA1c and its limitations
- ☒ Continuous glucose monitoring (CGM)
- ☐ AIDs use CGMs and algorithms to adjust insulin delivery. ^{ADA2026_ch7?}

Table 1.2: CGM-derived metrics presented in the ambulatory glucose profile report with target values.

Abbreviation	CGM-derived metric	Range	Recommended target
TIR	Time in Range	3.9–10.0 mmol/L(70–180 mg/dL)	> 70%
TBR	Time Below Range	< 3.9 mmol/L(< 70 mg/dL)	< 4%
TBR ₅₄	Time Below Range	< 3.0 mmol/L(< 54 mg/dL)	< 1%
TAR	Time Above Range	> 10.0 mmol/L(> 180 mg/dL)	< 25%
TAR ₂₅₀	Time Above Range	> 13.9 mmol/L(> 250 mg/dL)	< 5%
GMI	Glucose ManagementIndicator		No specific target
CV	Coefficient of Variation		≤ 36%

- [] Glycemic variability
 - Metrics of glycemic variability
 - Clinical challenges of glycemic variability (Think HbA1c and MBG and variability study)

1.6 Cardiovascular diseases and Diabetes

- Cardiovascular complications
- Heart failure in diabetes

Williams2020 expenditures and complications: <https://www.sciencedirect.com/science/article/pii/S0168822720301388> ## Cardiovascular risk prediction in diabetes * Traditional risk models * SCORE2-Diabetes risk score * What is SCORE2-Diabetes 10 year CVD risk score? * Limitations of conventional predictors

1.7 Machine learning

Relevant? Artificial Intelligence: The Future for Diabetes Care: <https://www.clinicalkey.com/#!/content/playContent/1-s2.0-S0002934320303399?returnurl=https:%2F%2Flinkinghub.elsevier.com>

- Data-driven vs hypothesis-drivenly(knowledge-based) paradigms
- Why machine learning in diabetes and cardiovascular risk prediction?
- Why are ML fantastic for CGM data?
- Types of machine learning methods
 - Logistic regression
 - LSTM and CNNs
 - Foundation models and their advantage
 - The CGMformer is bla bla bla¹³
- Challenges and pitfalls of machine learning in healthcare
 - Overfitting and generalizability
 - Interpretability and explainability

1.8 Fairness and bias in machine learning

- Define race and ethnicity
- Algorithmic bias and group fairness
- Fairness metrics and mitigation strategies
- Trends in the quality of care and racial disparities in Medicare managed care:
<https://pubmed.ncbi.nlm.nih.gov/16107622/>

Specific Aims 2

Aims {#sec-aims}

The overall aim of this PhD dissertation was to investigate the potential of continuous glucose monitoring (CGM) data to improve cardiovascular risk assessment and to evaluate the role of machine learning in extracting clinically relevant information from CGM data across different diabetes populations.

Aim 1: To identify and summarize the existing literature on the associations between glycemic control and cardiovascular outcomes, including the use of CGM-derived metrics for cardiovascular risk prediction. (*Study I*)

Aim 2: To determine whether a CGM foundation model captures meaningful and interpretable glucose representations in individuals with type 1 and type 2 diabetes. (*Study II*)

Aim 3: To evaluate whether conventional CGM-derived metrics and learned glucose representations improve prediction of elevated NT-proBNP beyond established clinical risk factors in individuals with type 2 diabetes. (*Study II*)

Aim 4: To evaluate the fairness of machine learning models for glucose prediction and compare strategies for tailoring prediction models to individual characteristics. (*Study III*)

3 Methods

In this chapter the three studies included in this PhD will be presented. The data will be described together with the chosen methods of the three studies and the different clinical variables, CGM-derived metrics and relevant cardiovascular outcomes will be explained.

- ☐ What was done (Description)
- ☐ Why it was done (Justification)
- ☐ tell about the 7-point fingerprick?

3.1 Overview of the Studies

Table 3.1: Overview of models and predictors

Model family	Model	Baseline predictors	Additional predictors
Clinical	A0	Age	–
Clinical	A1	Clinical variables*	–
Clinical	A2	Clinical variables*	TIR + MAGE
Clinical	A3	Clinical variables*	UMAP dimensions
SCORE2-Diabetes	B1	SCORE2-Diabetes score	–
SCORE2-Diabetes	B2	SCORE2-Diabetes score	TIR + MAGE
SCORE2-Diabetes	B3	SCORE2-Diabetes score	UMAP dimensions
CGM only	C1	TIR + MAGE	–
Embeddings only	D1	UMAP dimensions	–

* Clinical variables are the same as those included in the SCORE2-Diabetes risk model: age, sex, smoking status, systolic blood pressure, HDL cholesterol, total cholesterol, diabetes duration, HbA1c, and eGFR¹⁴.

TIR = time in range; MAGE = mean amplitude of glycemic excursions.

Table 3.2: Overview of the three studies included in this PhD

	Study I	Study II	Study III
Title	Continuous Glucose Monitoring-Derived Metrics and Cardiovascular Risk Among People With Diabetes: Systematic Scoping Review	Foundation model-derived representations of continuous glucose monitoring data: relation to glycemic patterns and cardiovascular risk	Racial disparities in continuous glucose monitoring-based 60-min glucose predictions among people with type 1 diabetes

	Study I	Study II	Study III
Study Type	Review	CGM foundation model evaluation study	Algorithmic fairness study
Data Sources	Literature on Embase and MEDLINE	The CANCAN study and the T1D Exchange (T1DX) dataset	T1D Exchange (T1DX) dataset
Population	Studies including people with type 1 diabetes, type 2 diabetes and/or prediabetes	159 people with type 2 diabetes 205 people with type 1 Diabetes	205 people with type 1 Diabetes
Input, predictors and exposures	Search strategy and eligibility criteria	Clinical variables, CGM-derived metrics and continuous CGM data	Continuous CGM data
Outcome, target	All studies that uses CGM data to say something about cardiovascular disease risk and outcomes	Visualisation of foundation model embeddings and elevated NT-proBNP prediction	Next blood glucose value
Analysis	Systematic scoping review, study characterization, and narrative synthesis	Foundation models and logistic regression models	Long short term memory models (LSTM) and Linear mixed effect models (LMEM)

3.2 Study I: A Scoping Review

A scoping review was conducted to investigate current knowledge of CGM-data and its possible connection and predictive power to clinical and subclinical cardiovascular outcomes. Given the expected heterogeneity between studies, a scoping review was considered the most appropriate methodology. The scoping review followed the Manual for Evidence Synthesis (Chapter 10 - Scoping reviews) from the Joanna Briggs Institute and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses extension for scoping review (PRISMA-ScR) guidelines.^{15,16}

3.2.1 Concepts and Definitions

The review focused solely on CGM-data for the input to any association or prediction models in the literature. This meant that studies only investigating self-monitored blood glucose with finger-prick measurements or blood sample measurements such as HbA1c were not considered in this review. CVD outcomes were divided into subclinical or clinical. For the clinical CVD, the following outcomes were considered: cardiovascular mortality, major adverse cardiovascular events, coronary artery disease, heart failure, stroke, and peripheral artery disease with synonymous terms included (e.g., ischemic heart diseases) and also clinical events (e.g., undergoing coronary artery bypass surgery). Subclinical CVD was grouped into six subcategories: arterial stiffness, flow resistance, arterial wall thickness, arterial wall composition, cardiac and pulse-related measures, and arterial lumen. CVD outcomes did not include broader risk factors nonspecific to cardiovascular risk (e.g., age or sex). Eligibility criterias for the scoping review¹⁷ (Study I) can be found in Table 3.3.

Table 3.3: Eligibility criteria for the scoping review (Study I)

Inclusion Criteria	Exclusion Criteria
Human clinical studies including participants with prediabetes or any type of diabetes except for gestational diabetes, regardless of definitions Peer-reviewed published original articles (including brief reports). Studies investigating either: a) The association between CGM-derived metrics of glycemic control and cardiovascular risk markers or cardiovascular diseases (prevalent or incident). b) CGM-derived metrics of glycemic control as predictors of cardiovascular risk markers or cardiovascular diseases (prevalent or incident).	Review articles, editorials, case reports, protocols, conference abstracts, and preprints. Animal studies not including any human participants. Studies not including metrics derived from continuous glucose monitoring device data. Studies including CGM-derived metrics as outcomes. Studies not focusing on CVD outcomes according to our definition, as outlined in the “Concepts and Definitions” section. Studies focusing on pregnant women with any form of diabetes including gestational diabetes. Studies where participants were monitored post-surgery or during hospitalization (e.g. Intensive care unit). Language not understood by authors

3.2.2 Search and Selection of Sources

The databases Embase and Medline were searched from inception to March 11, 2025. The search strategy was tested against 13 articles in the field, found through an initial exploratory search.^{18–30} The full search strategy can be found in Appendix A.

All identified records were uploaded into EPPI reviewer 6³¹ and duplicates were removed. Titles and abstracts were screened by at least two independent reviewers and a meeting was held after ~5% of the titles and abstracts had been screened, to resolve disagreements and ensure consistency between reviewers. Afterwards full-text screening was performed by two independent reviewers and held up against eligibility criteria. Disagreements were resolved through discussion at physical meetings between reviewers.

Backward and forward citation searching was performed on all the included studies after the screening process using citationchaser³² and the screen process was repeated until no further new studies were identified through the citationchaser.

3.2.3 Data Charting Process and Data Items

Research questions were defined before the screening process and data was extracted based on these research questions (Note 1) and following the requirements from the PRISMA-ScR checklist.¹⁵

Note 1: Research Questions (Study 1)

1. Is there an association between glycemic control and CVD risk?
2. Can CGM-derived metrics predict CVD risk?
3. What CGM-derived metrics are used in the literature?

4. Which cardiovascular markers and diseases are included as outcomes in the studies?
5. What characterizes study populations (age, sex, ethnicity, geographic location)?
6. What study designs are used (e.g., longitudinal cohort studies, randomized controlled trials, cross-sectional studies)?
7. How was data collected (e.g., clinical trial, epidemiological study, routinely collected data)?
8. What CGM devices were used?
9. What statistical models were used in the studies?
10. Is the data openly available?
11. Is the code openly available?

3.2.4 Synthesis of Results

Study characteristics were synthesized using descriptive statistics and supplemented by a narrative summary. Only the most adjusted models in the main population for each study was considered in this review. This approach was necessary as several studies reported numerous estimates across varying adjustments, different combinations of CGM-derived metrics and CVD outcomes, and/or distinct subgroup stratifications. Given the large volume of literature, studies investigating clinical cardiovascular outcomes were prioritized in the main synthesis. While subclinical markers offer valuable insight into early vascular changes and disease progression, clinical outcomes are more directly relevant to cardiovascular risk prediction and clinical decision-making. The main synthesis of clinical outcomes included both effect estimates and measures of statistical significance whereas the supplementary synthesis on subclinical CVD focused primarily on creating an overview of the chosen CGM-derived metrics and the statistical significance of reported associations. All evaluation metrics mentioned in the studies doing prediction analyses were reported based on what the studies reported.

3.3 Study Populations

Two datasets were used in this PhD (Study II and study III) which are presented in the following sections

3.3.1 CANCAN

The CANCAN dataset was collected in an observational case-control study from three hospital outpatient clinics (Denmark). The aim of the CANCAN Study was to investigate if cardiac autonomic neuropathy (CAN) assessments, CGM and heart failure indicators could be used in risk stratification in people with T2D. Participants were included if they were adults (< 18 years old) and had T2D for at least a year. Exclusion criteria were atrial fibrillation or laser-treated eye disease within the past three months, were pregnant or lactating, had a life-threatening disease (expected survival < 1 year), or had cognitive disabilities rendering them unable to understand and give consent. Data were collected during a clinical examination, during which blood and urine samples were obtained, and a blinded CGM sensor was inserted. Blood samples to obtain NT-proBNP values were collected when the CGM sensors were returned, after the 2-week CGM data collection period had ended (Figure 3.1). In total, 159 participants were included since the rest were missing age, sex, HbA1c, BMI, NT-proBNP, HDL, total cholesterol, or eGFR and these values were needed for the analyses in Study II.

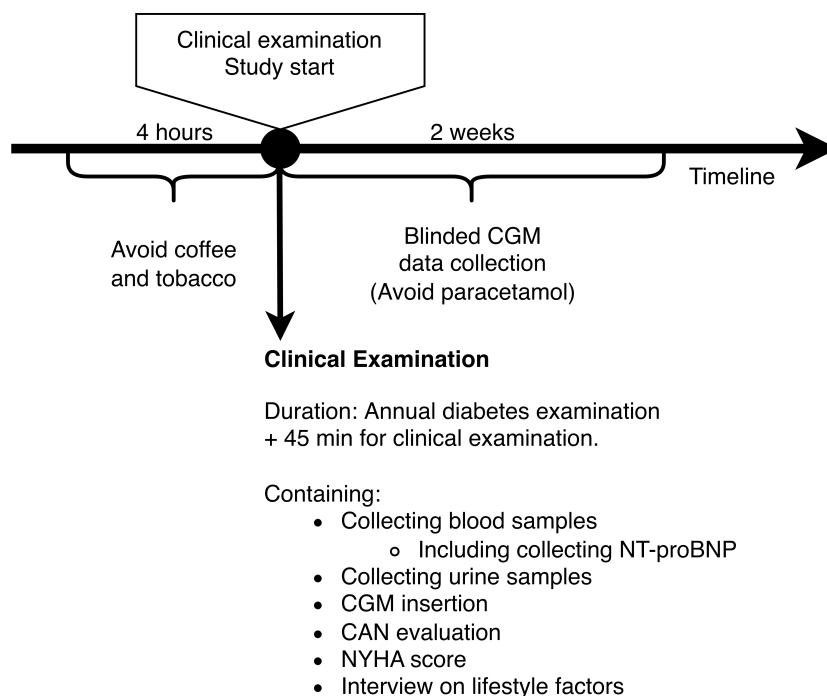


Figure 3.1: Data collection timeline for the CANCAN Study

3.3.2 T1DX

The publicly available T1D Exchange (T1DX) dataset, distributed by the JAEB Center for Health Research under the protocol “Racial Differences in Mean CGM Glucose in Relation to HbA1c, was collected in 10 diabetes centers (US).³³ The data was collected to determine if there is a racial difference in the association between mean glucose and glycated hemoglobin (HbA1c) between non-Hispanic African Americans or black (Black) and non-Hispanic white (White) people with T1D. In the T1DX study ethnicity and race was defined through self-identified and self-reported ethnicity/race verification forms where ethnicity was defined as Hispanic or Latino as: “a person of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of race. The term “Spanish origin” could be used in addition to “Hispanic” or “Latino” “.³⁴ For race a participant defined as White was “a person having origins in any of the original peoples of Europe, the Middle East or North Africa.” and Black or African American as “a person having origins in any of the Black racial groups of Africa. Terms such as “Haitian” or “Negro” can be used in addition to “Black” or “African American” “.³⁴ In this thesis, the term race will be used moving forward, as it aligns with the terminology used by Bergenstal et al.³⁴, it is, however, acknowledge that it is inherently difficult to define race.³⁵ The T1DX study adhered to the tenets of the Declaration of Helsinki and was approved by the local institutional review boards and 205 people completed the study by providing written informed consent and assent when applicable.³⁴ All 205 participants were included in further analyses. Of the 205 participants, 101 were White and 104 were Black. For each race, two age groups were sampled 8 to <18 years old and >18. Each participant wore a blinded modified investigational version of Abbott Diabetes Care’s Flash Glucose Monitoring System (CGM) as continuously as possible for 12 weeks. CGM data was sampled every 15 minutes in a range between 2.21 - 27.65 mmol/L (40 and 498 mg/dL). Blood samples were taken for measurement at enrollment, at some of the follow-up visits, and at the end of enrollment.

3.4 Study II: Foundation Models and Predictive Utility of CGM data

The second study was focused on investigating the knowledge and predictive utility of CGM data, both extracting a novel CGM-derived metric and the classical known ones found in study I.¹⁷

3.4.1 Study II: Data and Participant Selection

Both the CANCAN and the T1DX dataset was used for Study II. Participants with missing age, sex, HbA1c or BMI were removed for both cohorts and additional CANCAN participants with missing NT-proBNP, HDL, total cholesterol, and eGFR were also removed, since it was wished to calculate the SCORE2-Diabetes risk score. T1DX participants with missing race data were also removed. Days with less than 70% of CGM data were removed as well as participants with less than one day of data from both datasets to ensure sufficient data for the analyses. The final number of participants included in Study II was 159 from the CANCAN dataset and 205 from the T1DX dataset.

3.4.2 Embeddings

Two promising CGM specific foundation models exist, both published within the last year.³⁶ Of the two foundation models, only one model is fully open access, the CGMformer.¹³ The CGMformer was therefore chosen and the data was organized in 24-hour long-format input to match the input requirements of the CGMformer. The output of the foundation model is an embedding, which is a numerical vector that summarizes the characteristics of the input glucose profile. A simple vector can be drawn as an arrow in a 2-dimensional space as shown in Figure 3.2, however the CGMformer gives a 128-dimensional embedding vector for each 24-hour CGM data input which is not possible to visualize. Uniform Manifold Approximation and Projection (UMAP) algorithm was used to reduce the dimensionality of the embeddings to two dimensions for visualization purposes and to compare with current literature.^{13,36,37}. The UMAPs were colored by TIR, and MAGE to investigate if the embeddings were able to capture relevant information about glycemic patterns. History of CVD and elevated NT-proBNP for the CANCAN embeddings were also used to color the UMAPs to investigate if the embeddings were able to capture relevant information about cardiovascular risk and race was used to color the T1DX embeddings to investigate if the embeddings were able to capture relevant information about racial differences in glycemic patterns.

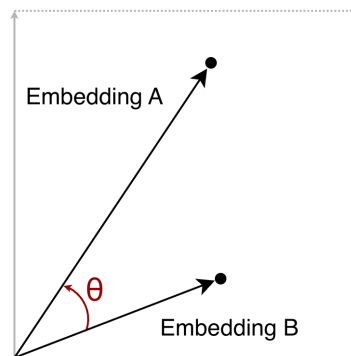


Figure 3.2: Embeddings shown as 2-dimensional vectors with the angle θ between the two embeddings.

Because embeddings are numerical vectors, they can be compared to quantify the similarity between glucose profiles. Cosine similarity was used because it measures the angle between

two embedding vectors independent of their magnitude (the length of the vectors). The cosine similarity is calculated by measuring the cosine angle between two vectors as shown by θ in Figure 3.2. The cosine similarity is a number between -1 and 1, where 1 means the two vectors are identical, 0 means they are unrelated, and -1 means that they are pointing in opposite directions (Figure 3.3). For the analyses in this study, cosine distance ($1 - \text{cosine similarity}$) was used because it expresses similarity as a distance measure, allowing embeddings with similar orientations to have small distances and facilitating downstream analyses and visualizations. The cosine distances range from 0 to 2 where 0 means the two vectors are identical, 1 means they are unrelated, and 2 means that they are pointing in opposite directions.

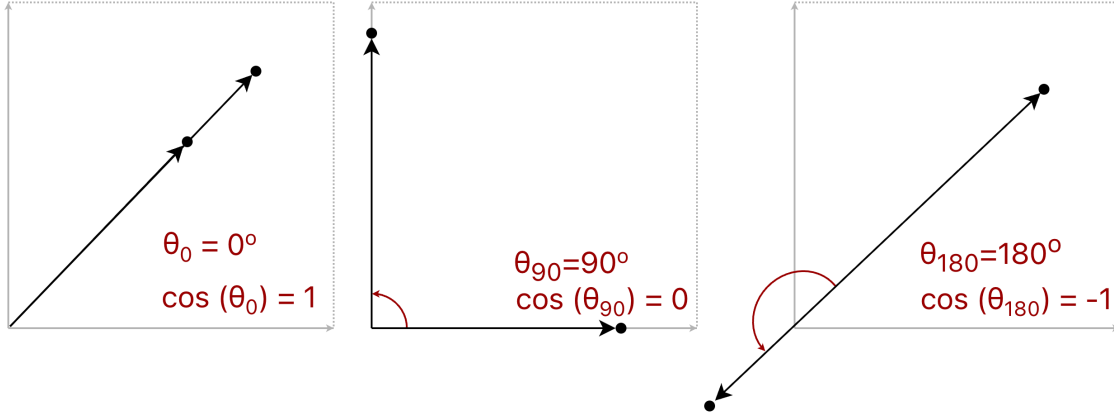


Figure 3.3: Embeddings shown with different cosine similarities. The cosine similarity is calculated as the cosine of the angle (θ) between two embeddings.

Kernel density plots were used to visualize the distribution of cosine distances within specific clinical strata and across differing participants (Figure 3.4). Groupings were made based on days from the same participant, days from different participants with the same clinical strata, and days from different participants with different clinical strata. The main focus was on TIR and HbA1c were the thresholds were as followed: TIR 70% (70–180 mg/dL, 3.9–10.0 mmol/L) and HbA1c $<7.0\%$ (< 53 mmol/mol). Other UMAPS coloring and groupings were also explored including race, MAGE, previous history of CVD, NT-proBNP, and SCORE2-Diabetes scores.

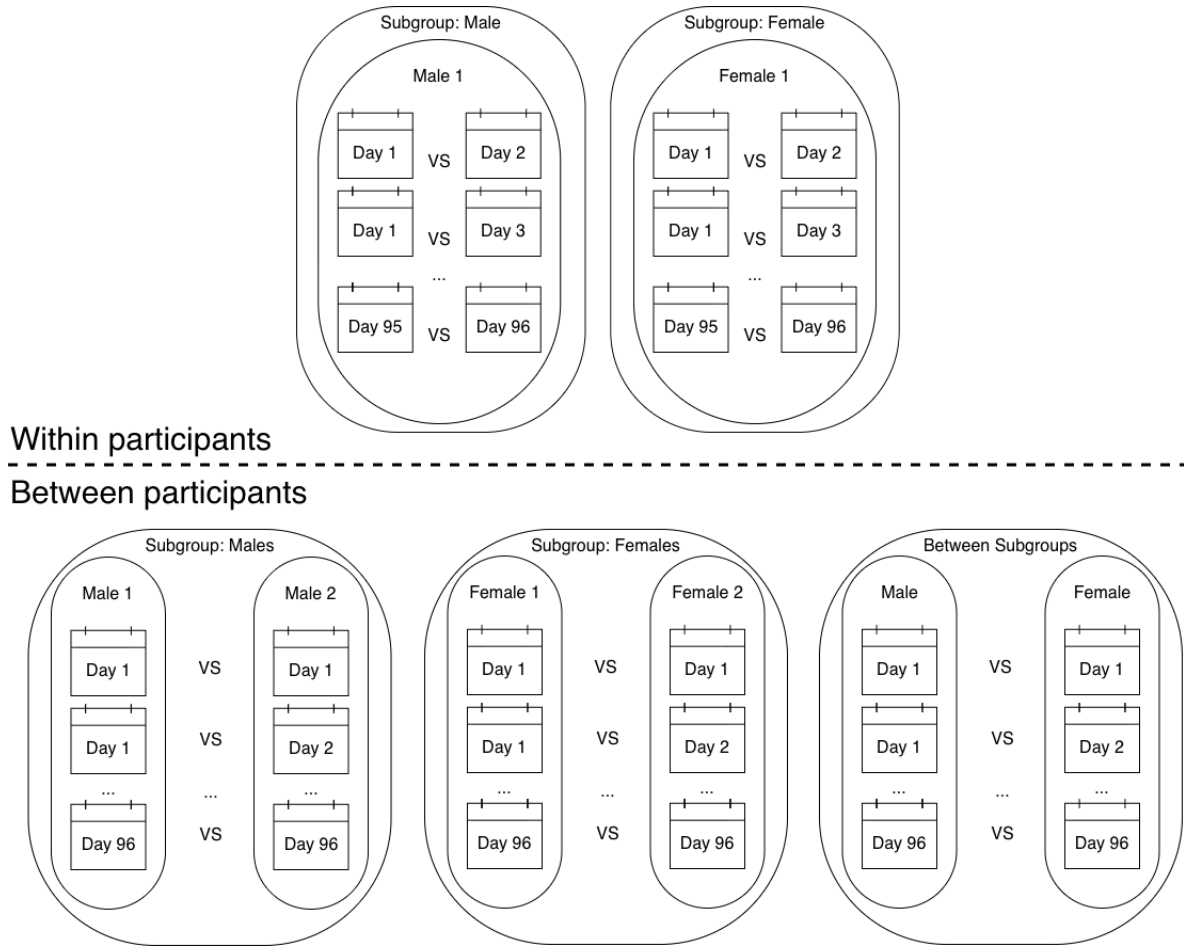


Figure 3.4: The sex strata is used as an example to illustrate the cosine distance analysis of the within and between cosine distance pairs

3.5 Study II: Cardiovascular Risk Assessment

The following section is about cardiovascular risk assessment evaluated predicting NT-proBNP as a binary outcome using a >125 pg/mL threshold.^{source?} The CANCAN dataset was the only dataset used for this CVD risk assessment. The focus was to test the predictive power of CGM data both as represented as embeddings and as well-established CGM-derived metrics. Some models also included clinical predictors to compare up against baseline models.^{source_kovatchev?orAnotherSourceOnMAGE?} Prediction models integrated clinical, standard CGM-derived, and learned representations.

3.5.1 NT-proBNP - The target

- ☐ What is NT-proBNP (scale, unit and connection with heart failure)

3.5.2 Predictors

- ☐ What is the SCORE2-diabetes and why was it chosen? (The Outcome + european dataset) Inspiration was drawn from the SCORE2-Diabetes risk score, which is a 10-year CVD risk score for european people with diabetes. It was decided to use the input

predictors of the SCORE2-Diabetes risk score as the clinical set of predictors used in this study.^{source?} This resulted in set of clinical predictors including age, sex, smoking status, systolic blood pressure, total cholesterol, and HbA1c.^{source?} Initial testing quickly showed that the numbers of predictors had to be limited and the SCORE2-Diabetes risk score was chosen as another way to represent the clinical predictor set while keeping the number of predictors down. Additionally age was tested as a predictor on its own to see if it was a strong predictor of elevated NT-proBNP, since NT-proBNP is known to increase with age.³⁸ Time in range (TIR) and mean amplitude of glycemic excursions (MAGE) were chosen as they were the CGM-derived metrics most commonly used in the literature based on the findings from Study I.¹⁷ All CGM-derived metrics in this thesis were calculated using the IGLU package^{source?} in R with default settings. TIR was calculated as the percentage of time spent in the range of 3.9–10.0 mmol/L (70–180 mg/dL).¹¹ MAGE was calculated as the mean amplitude of glucose excursions, where the included excursions were consecutive peak-to-nadir (or nadir-to-peak) glucose differences exceeding one standard deviation of the individual's glucose measurements. The iglu implementation computationally reproduces the original manual procedure proposed by Service et al.³⁹ To aggregate multiple days into a single observation per participant, daily CGMformer representations were minimum-, maximum-, and mean-pooled, and reduced to two dimensions via UMAP. Maximum pooling was considered the default based on previous work.³⁶ The CGMformer generated one 128-dimensional embedding for each 24-hour CGM recording. Because NT-proBNP was measured at the participant level, a single embedding was required for each participant. Therefore, embeddings from multiple recording days were aggregated using pooling to obtain one participant-level embedding. Both minimum, maximum, and mean pooling were explored to determine the most effective aggregation method. The resulting 128-dimensional embeddings were subsequently reduced to two dimensions using UMAP, thereby enabling their use in the prediction models while keeping the number of predictors down to a minimum.

3.5.3 Model development and evaluation

Prediction models were developed using logistic regression to predict elevated NT-proBNP (>125 pg/mL) as a binary outcome. The models were evaluated using the area under the receiver operating characteristic curve (AUROC) as the primary metric. Bootstrap with optimism correction was employed for internal validation because of the limited sample size and the absence of an external validation cohort. The method repeatedly sampled bootstrap samples by sampling with replacement from the original dataset. For each bootstrap sample, the prediction model is fitted and its performance is evaluated both in the bootstrap sample (apparent performance) and in the original dataset (test performance). The difference between these two performance estimates represents the optimism due to overfitting. Averaging the optimism across all bootstrap samples and subtracting it from the apparent performance of the model fitted to the full dataset yields an optimism-corrected estimate of model performance. Internal validation was performed using 500 bootstrap resamples.^{40,41} Risk distribution plots were reported for each model across all bootstrap samples.^{41,42}

3.5.4 Prediction scheme overview

The prediction models were structured hierarchically based on clinical variables (Model A) and the calculated SCORE2-Diabetes score (Model B) in the configurations shown in `?@tbl-models`

Table 3.4

Model family	Model	Baseline predictors	Additional predictors
Clinical	A0	Age	–
Clinical	A1	Clinical variables*	–
Clinical	A2	Clinical variables*	TIR + MAGE
Clinical	A3	Clinical variables*	UMAP dimensions
SCORE2-Diabetes	B1	SCORE2-Diabetes score	–
SCORE2-Diabetes	B2	SCORE2-Diabetes score	TIR + MAGE
SCORE2-Diabetes	B3	SCORE2-Diabetes score	UMAP dimensions
CGM only	C1	TIR + MAGE	–
Embeddings only	D1	UMAP dimensions	–

Overview of models and predictors {#tbl-models}

* *Clinical variables are the same as those included in the SCORE2-Diabetes risk model: Age, sex, smoking status, systolic blood pressure, HDL cholesterol, total cholesterol, diabetes duration, HbA1c, and eGFR.*¹⁴

TIR = Time in Range; MAGE = Mean Amplitude of Glycemic Excursions.

3.6 Study III: Algorithmic Fairness in Blood Glucose Forecasting

The third study was investigating racial bias in a data driven machine learning model only based on blood glucose measurements from a CGM-sensor as this is an important consideration in the path to get ML models into use

3.6.1 Preprocessing of CGM data for Forecasting

All data was sorted into 60 minutes windows with 4 values to predict 60-minutes into the future as visualized in Figure 3.5. There are no clear answers in the literature about which prediction horizon is the best to choose, however CGM devices measure in the interstitial fluid and there is a physiological delay between the measured blood glucose and the interstitial glucose levels.^{source?} Of the European approved CGM devices it is only the Dexcom that shows trends to the users and since CGM devices mainly focuses on monitoring, the forecasting question is only really interesting when administering insulin in order to catch hypo- and hyperglycemic events ahead of time.^{source?} It is very difficult to figure out how the prediction algorithms in insulin pumps are made and so far I have had very little luck in figuring it out. It is assumed that the companies use linear regression and the last 20 minutes of data to predict the next values which is usually the last four samples since the resolution of the used CGM devices usually are between 1 to 5 minutes.^{source?} With a CGM measurement every 15 minutes and the physiological delay in mind, the 60 minutes were decided to be a good prediction horizon. All samples with missing values were removed.

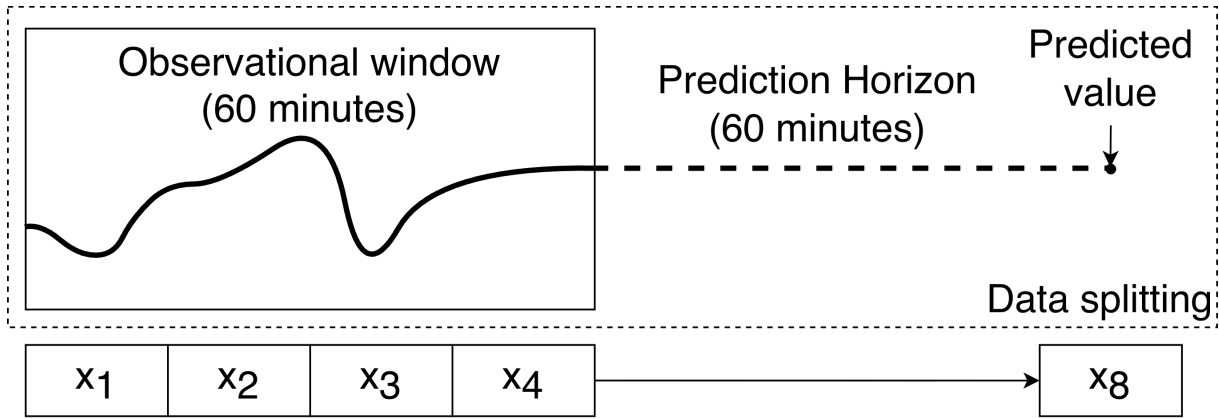


Figure 3.5: The prediction horizon and overview of the creation of samples (predictors and targets).

3.6.2 Partitioning of the Dataset

The T1DX dataset was randomly resampled without replacement at the participant level to avoid data leakage and to evaluate model performance on independent participants. As illustrated in Figure 3.6, training subsets with varying proportions of Black and White participants were generated by randomly selecting participants from the original dataset while keeping the train-test split at the participant level. Eleven training set compositions were created, ranging from 100% White/0% Black to 0% White/100% Black in 10% increments.

Because some training subsets consisted entirely of participants from a single racial group, the maximum training set size was constrained by the size of the smaller racial group minus the participant kept for testing. Consequently, each training subset contained approximately half the number of participants in the original T1DX dataset. Repeating this procedure for each participant resulted in 2,255 training datasets ($205 \text{ participants} \times 11 \text{ training set compositions}$) and 205 independent test sets. Throughout all resampling procedures, all CGM recordings from a participant were kept together to prevent data leakage between the training and test sets.

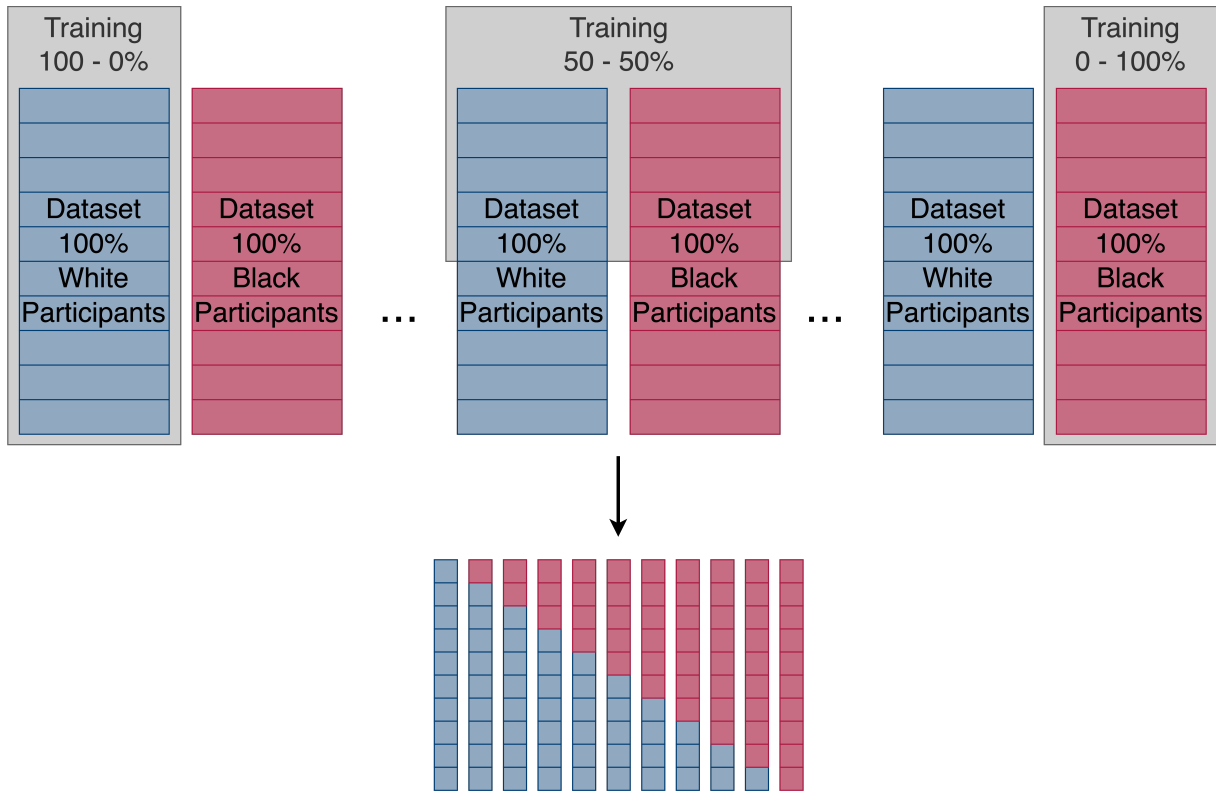


Figure 3.6: The prediction horizon and overview of the creation of subsets for training.

Eleven subsets were created for each of the participants and the last weeks worth of data was chosen to test on as illustrated in Figure 3.7. The one week was chosen because participants had varying amount of CGM data and the participant with the least amount of data had ~ 2 weeks, and all participants were wished to be kept due to the small number of participants in the T1DX dataset. The remaining data were used for training the models.

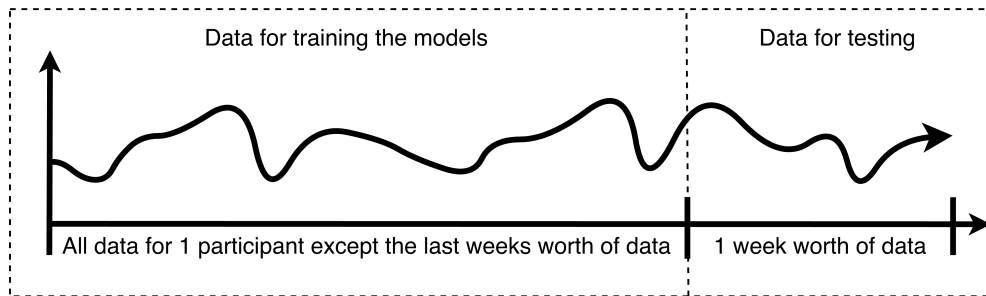


Figure 3.7: The splitting of a participants data to create training, fine-tuning and test sets

3.6.3 Training Approaches and Model Selection

Four training approaches were developed as illustrated in Figure 3.8. The training approaches included a naive baseline and three different machine learning models as described in [?@tbl-study3_training_approaches](#).

A long short-term memory model (LSTM) and overall architecture was chosen for the machine learning model, based on a previously described LSTM in van Doorn et al. [@vanDoorn2021](#) (two LSTM layers followed by a dense layer) since there were good performance with this model. All the training data were scaled to be within the range of zero to one. Hyperparameters included the number of neurons in LSTM layers, activation functions, dropout rate, batch size, learning rate, decay rate, and early stopping. These hyperparameters were tuned using a grid search,

heatmaps, and manual tuning. Further information is given in Thomsen et al.⁴³ and the code repository⁴⁴.

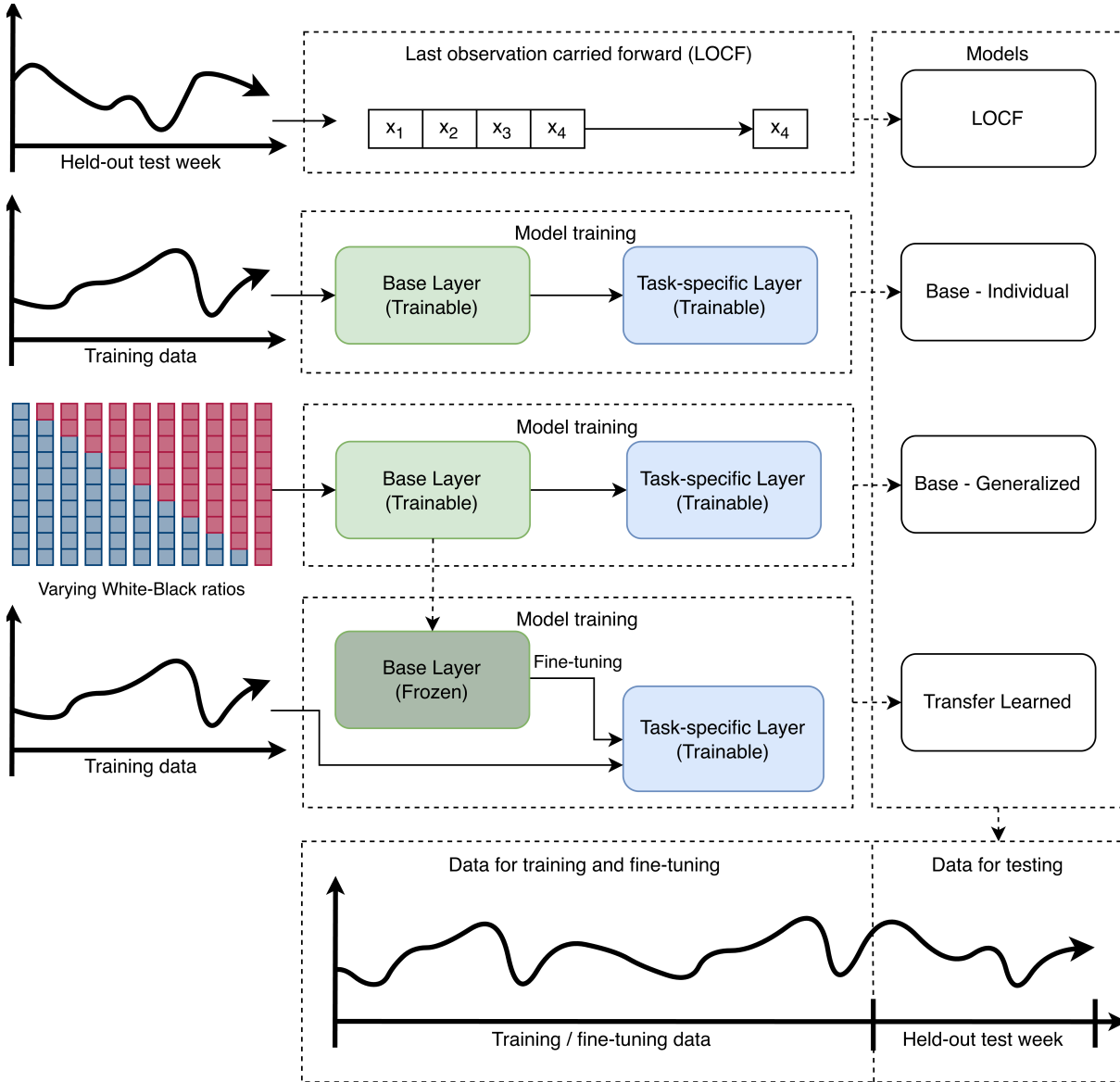


Figure 3.8: The four training approaches. The Base-Individual, Base-Generalized, and Transfer Learned models were based on an LSTM architecture. The base layer (green) consisted of two LSTM layers, while the task-specific layer (blue) consisted of a single dense layer.

One of the machine learning models were developed using transfer learning. It is a machine learning technique in which knowledge learned from one domain is transferred to another related domain. One common transfer learning approach is fine-tuning, where a pretrained model is further trained using new data.⁴⁵ In this analysis, generalized glucose prediction models were first trained using data from multiple participants (Base-Generalized model) and subsequently fine-tuned using participant-specific data to personalize glucose predictions (Transfer Learned model).

Model names	Description	Training data
Last observation carried forward (LOCF)	naive baseline: In the LOCF model, a given CGM measurement (x4) was used to predict blood glucose values 60 minutes in advance (x8) in the test set by assuming no change in the last 60 minutes.	No training data required
Base-Individual	In the Base-Individual model, each participant's complete CGM data (except for the last week, which was used for the testing data) was used for training.	Only data from 1 participant Figure 3.7
Base-Generalized	Here, for each participant, the 11 training corresponding datasets described above were used to generate 11 separate models, which were all subsequently tested on the given participant's last week of CGM data.	All the different subsets of data Figure 3.6
Transfer Learned	This model mirrored the Base-Generalized Model's setup. However, the 11 trained models for each participant were subsequently tailored, i.e., fine-tuned with transfer learning to the given participant's complete CGM data (except for the last week, which was used for the testing data)	Only data from 1 participant Figure 3.7

The four training approaches used to develop glucose prediction models for each participant in the T1DX dataset. `{#tbl-study3_training_approaches tbl-colwidths="[20,60,20]"}`

3.6.4 Model Evaluation

To evaluate model performance, each participant's last weeks worth of data (672 measurements) was reserved for internal testing (Figure 3.7). Model accuracy was assessed by calculating the root mean squared error (RMSE) for each participant across all four modeling approaches. For the LOCF and Base-Individual models, this yielded one RMSE value for each of the 101 White participants and 104 Black participants. For the Base-Generalized and Transfer Learned models, RMSEs were calculated for each participant across 11 data subsets, each representing a unique proportion of White and Black participants. Performance was summarized as the mean RMSE with corresponding 95% confidence intervals (CIs) for each racial group.

To evaluate the predictive performance across different ratios for the Base-Generalized and Transfer Learned models a linear mixed effect model (LMEM) was used since data was dependent due to participants having an RMSE value for each subset with varying proportions of White and Black participants. It was assumed that RMSE values would vary between individuals and therefore a random intercept was included and the following covariates were included: ratio (proportion of White and Black), race, age, sex, and training data sample size. (Categorical: Race, age, and sex; continuous: RMSE, ratio, and sample size) It was further assumed that the varying proportions of Black and White participants in the training data subsets would affect the performance and an interaction term was therefore added. The interaction term allowed for different slopes in the LMEM which allowed us to test the hypothesis. It was expected that when plotting the proportion of White participants in the training data against the RMSE, the slope would be negative for White participants and positive for Black participants (Figure 3.9). This would indicate that as the proportion of White participants in the training data increased, the RMSE decreased for White participants and increased for Black participants, indicating racial bias in the model. The Base-Individual model would perform the worst since it was trained on the least amount of data and the Transfer Learned model would remove the slopes that were expected to be seen in the Base-Generalized models.

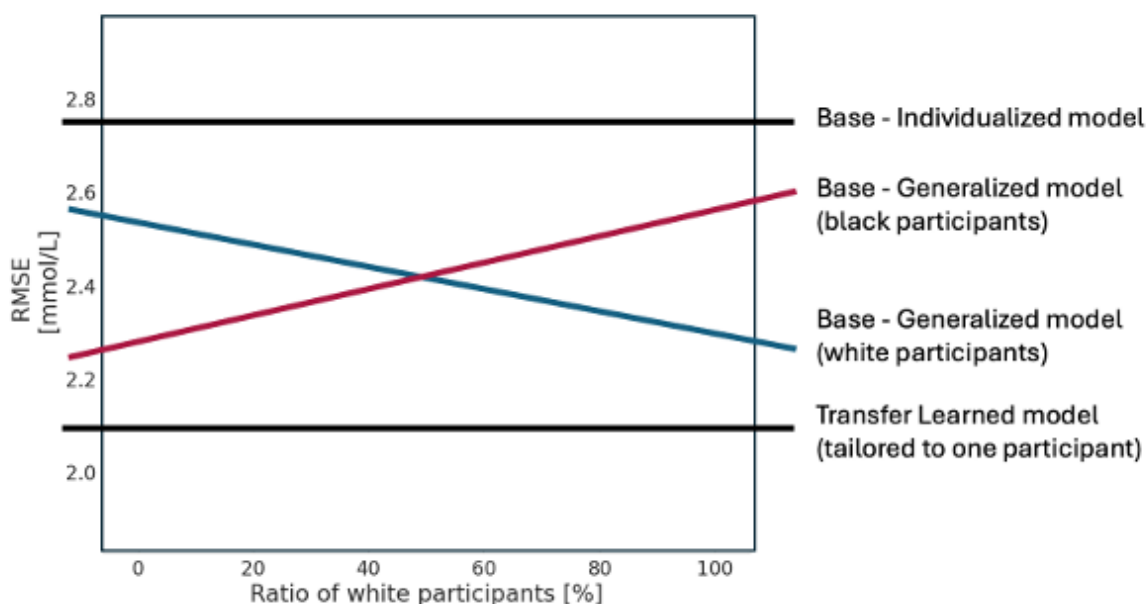


Figure 3.9: The hypothesis for the results of the linear mixed effect models.

Surveillance error grids (SEG) were used to get a general idea of the clinical risk of the prediction models.⁴⁶ The error grids are originally developed for the regulatory bodies to assess clinical accuracy of blood glucose monitoring in post-surveillance before the wide spread use of CGM devices.⁴⁷ and the results should therefore be interpreted with caution when using CGM data. The SEG is a 600x600 grid assigned an individual risk score to each datapoint with 15 colored zones, each indicating severity of a monitoring error that is acted upon in regards to risk of hypoglycemia or hyperglycemia (Figure 3.10).⁴⁷ To facilitate interpretation, the 15 surveillance error grid zones were collapsed into the five broader risk zones (“No”, “Slight”, “Moderate”, “High”, and “Extreme”), as previously suggested in the literature.⁴⁶ Predictions outside of 0–33 mmol/L were changed to 0 or 33 mmol/L to keep within the upper and lower bounds of the surveillance error grid.

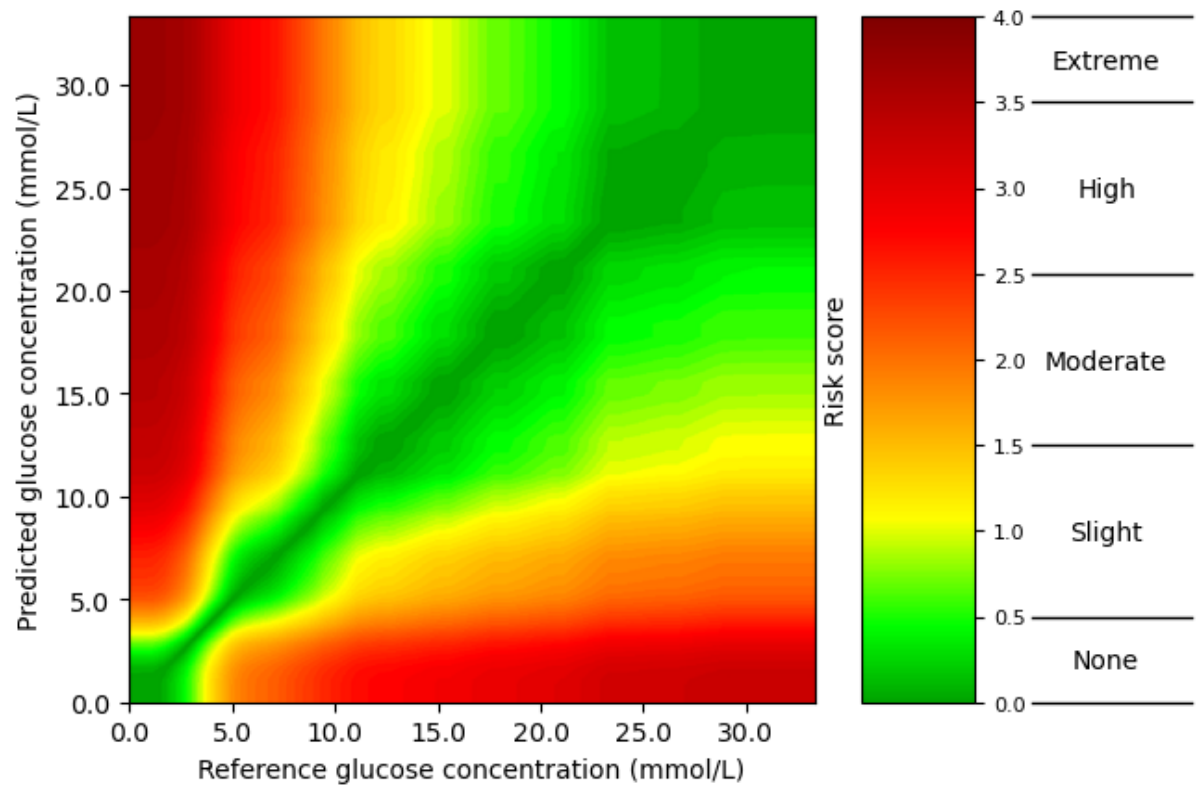


Figure 3.10: An empty surveillance error grid

4 Results

4.1 Study I: Results from a Scoping Review

4.1.1 Study Selection

The study selection process is summarized in Figure 4.1. The Medline and Embase search resulted in 4884 records screened on title and abstract, and 81 reports were assessed for eligibility based on full-text review by two independent reviewers. The citation search on the included 51 reports resulted in 2 additional reports, which finally resulted in 53 included reports in the review. All included and excluded reports are reported in Appendix B.

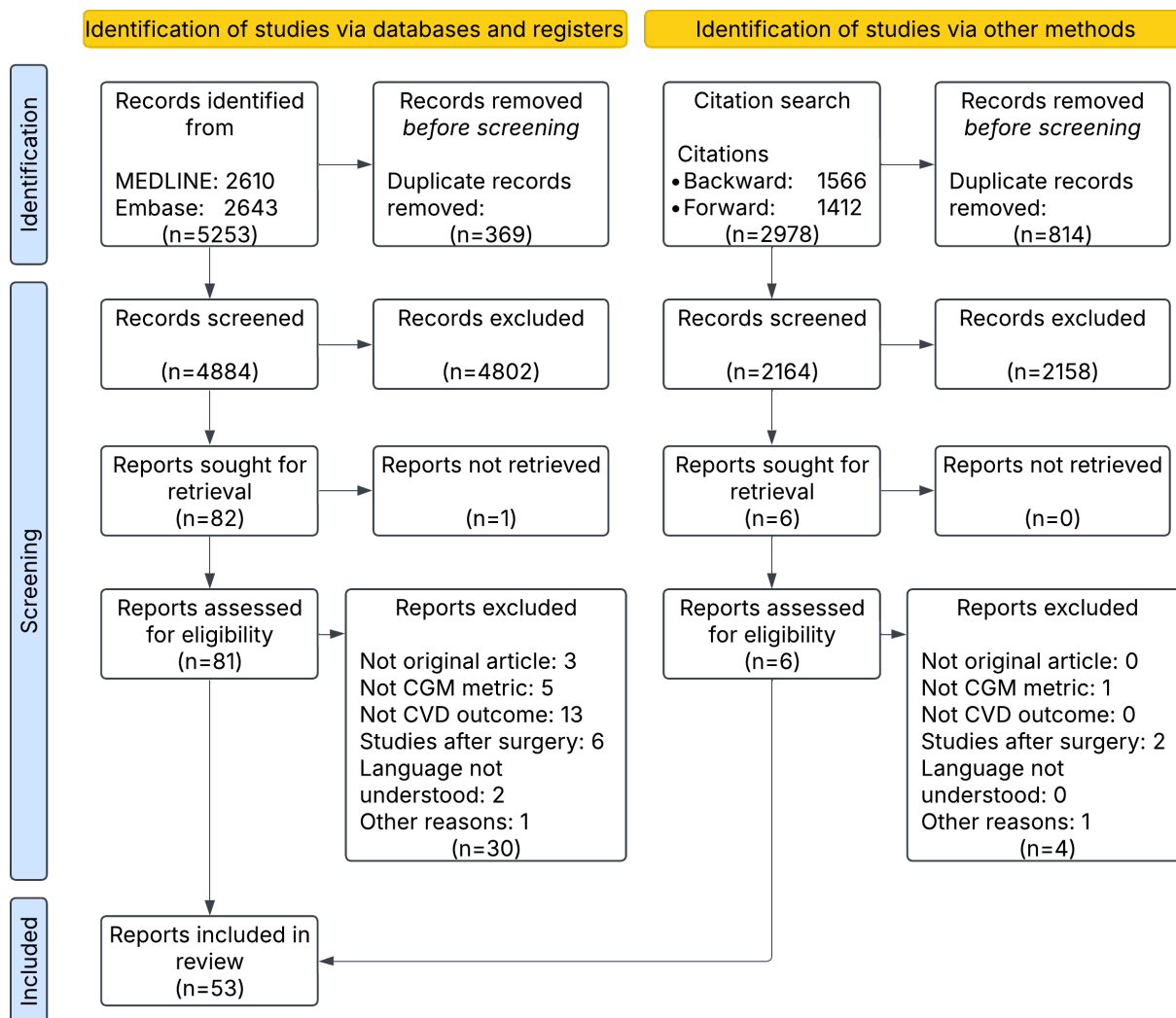


Figure 4.1: Flow diagram of study selection

4.1.2 Study Populations

The study population most common was people with T2D (34 of studies), then people with T1D (20 of 53) and lastly prediabetes (4 of 53). Most studies included adult participants, whereas seven focused on children younger than 18 years. Two studies included only male participants, whereas the remaining studies included both male and female participants.

The geographic distribution was found based on the hospitals and centers where data was collected or by the first author's affiliation when the other option was not available. The distribution was uneven as shown in Figure 4.2. Most studies were conducted in Asia (27 of 53) or Europe (22 of 53). Three studies included data from North America, one was conducted in Australia, and only one included data from an African country, although its study population was predominantly White.^{Borg2011?} No data was included from South America (Appendix C).

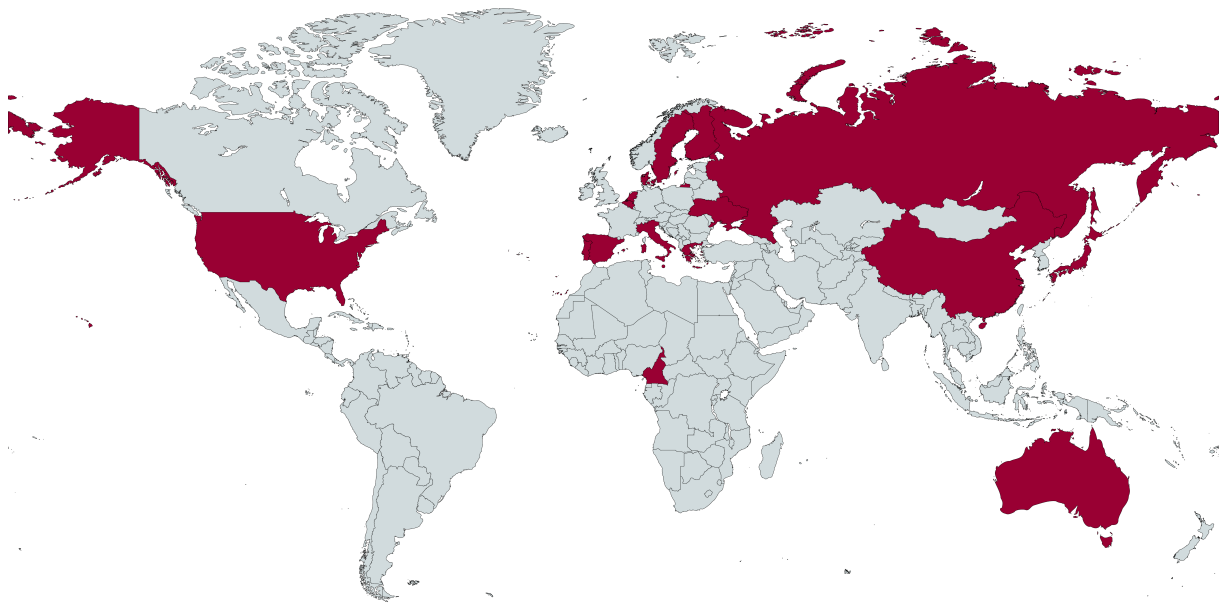


Figure 4.2: Geographical location of the studies. Countries colored in red indicate the geographical locations where the studies were conducted. Map was created using MapChart <https://www.mapchart.net/>.

4.1.3 Study Designs and Demographics

Only one study was a randomized controlled trial,^{33?} and the remaining studies were observational. Most studies used a cross-sectional design, with the cardiovascular outcome assessed at the same time as the CGM sensor was given. Only six studies performed longitudinal analyses. Study sample sizes varied considerably, ranging from 17^{34?} to 6,225^{18?} participants, with a median of 152. CGM data were obtained either from participants' own devices (11 studies) or from CGM devices provided as part of the study (42 studies). Statistical analyses most commonly included linear regression (22 studies) and logistic regression (22 studies), followed by Spearman or Pearson correlation analyses (11 studies) and Cox proportional hazards regression (3 studies). Several studies also performed additional exploratory analyses, such as correlation analyses, to identify relevant covariates for inclusion in regression models.

All included studies primarily investigated associations between CGM-derived measures and cardiovascular outcomes. Only three studies reported predictive performance measures as part of secondary analyses,^{Koroleva?} and none applied machine learning methods. Furthermore, none of the included studies provided openly available data or source code. Instead, availability was either not addressed or limited to access upon reasonable request from the corresponding author.

4.1.4 Continuous Glucose Monitoring

Medtronic CGM devices were the most frequently used (28 of 53 studies), followed by Abbott (9 of 53) and Dexcom (5 of 53). Other manufacturers included Menarini, Meiqi Company, and SIBIONICS. Sampling intervals ranged from 3 to 15 minutes but were reported in only a minority of studies (21 of 53). TIR (22 studies) and MAGE (23 studies) were the most frequently investigated CGM-derived metrics (Table 4.1) When stratified by T1D and T2D then TIR was the most frequently investigated metric in T1D populations (10 of 23 studies), whereas MAGE was less commonly investigated (5 of 22 studies). In T2D populations, TIR and MAGE were investigated in 11 of 23 and 14 of 22 studies, respectively. Studies conducted in T2D populations also tended to include larger sample sizes than those in T1D populations (median [IQR]: TIR, 510 [405–600] vs. 214 [119–547]; MAGE, 251 [89–411] vs. 57 [30–215]).

Table 4.1: Summary of the popularity of the seven most used CGM-derived metrics

CGM-derived Metrics:	TIR	MAGE	MBG	SD	CV	TBR	TAR
Number of Studies, n	23	22	21	19	19	16	15
T1D, n	10	5	9	7	8	9	8
T2D, n	11	14	8	10	8	7	7
Mixed*	2	3	4	2	3	0	0
Total study size for the CGM-derived metrics**	445	156	105	267	342	123	152
T1D**	214	57	105	267	152	124	138
T2D**	510	251	405	371	499	121	405
	[157-577]	[63-344]	[71-427]	[77-480]	[98-577]	[77-312]	[77-425]
	[119-547]	[30-215]	[31-347]	[54-515]	[91-515]	[102-161]	[98-188]
	[405-600]	[89-411]	[77-499]	[95-484]	[358-632]	[75-499]	[75-499]

* Mixed includes both T1D, T2D and prediabetes

** Mixed includes both T1D, T2D and prediabetes

Abbreviations: T1D: type 1 diabetes, T2D: type 2 diabetes, MAGE: mean amplitude of glycemic excursions, TIR: time in range, MBG: mean blood glucose, SD: standard deviation, CV: coefficient of variation, TBR: time below range, TAR: time above range.

4.1.5 Prediction studies

Only three studies evaluated the predictive performance of CGM-derived metrics for cardiovascular outcomes, all using logistic regression models with MAGE as the predictor. Two studies reported receiver operating characteristic area under the curve (ROC-AUC) values of 0.61 and 0.62, respectively,^{43?} and both found MAGE to outperform HbA1c (ROC-AUC: 0.55 and 0.58, respectively). The third study used receiver operating characteristic analysis to determine the optimal threshold for dichotomizing MAGE during variable selection but did not report the ROC-AUC.^{44?} Instead, a threshold of MAGE ≥ 3.4 mmol/L was identified as a risk factor for stenosis and/or occlusion, with a sensitivity of 0.60 and a specificity of 0.61.

4.1.6 Association studies

An overview of the different outcomes found in the literature both clinical and subclinical are shown in Table 4.2.

Table 4.2: Summary of established clinical and subclinical outcomes and corresponding abbreviations identified in the literature.

Clinical	Subclinical
CVD = Cardiovascular disease	ABI = Ankle-brachial index
- Macrovascular complications	BP = Blood pressure
- Diabetic macroangiopathy	Carotid artery distensibility = Carotid artery distensibility or carotid distensibility coefficient
CAD = Coronary artery disease	CCA = Common carotid artery
- CHD = Coronary heart disease	CIMT = Carotid artery intima-media thickness
- MI = Myocardial infarction	DBP = Diastolic blood pressure
- Heart attack	FMD = Flow-mediated dilation of the brachial artery
- Heart failure	GSM = Gray-scale median of the carotid arteries
- (Unstable) angina pectoris	HRV = Heart rate variability
Cardiovascular events	IMT = Intima-media thickness
- Ischemic heart disease (IHD)	PWV = Pulse wave velocity
LEAD = Lower extremity artery disease	- ba-PWV = Brachial-ankle pulse wave velocity
MACE = Major adverse cardiovascular events	- cr-PWV = Carotid-radial pulse wave velocity
PAD = Peripheral artery disease	- cf-PWV = Carotid-femoral pulse wave velocity
Stroke = Stroke, Cerebrovascular disease, or Cerebrovascular accident	- Aortic PWV = Aortic pulse wave velocity
	SBP = Systolic blood pressure

4.1.7 Clinical Association Studies

Most studies focused exclusively on subclinical cardiovascular outcomes (37 of 53 studies, 70%), whereas 13 studies (25%) investigated only clinical cardiovascular outcomes and three studies (6%) included both clinical and subclinical outcomes. Because the subsequent chapters of this dissertation focus on clinical cardiovascular outcomes, only these findings are discussed in detail below (Table 4.3 and Table 4.4). A summary of studies investigating subclinical cardiovascular outcomes is provided in Appendix D.

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Chen, 2020 [†]	Longitudinal (prospective) and cross-sectional (retrospective) FU: in-hospital or within 3 months after discharge from hospital	T2D with CAD (Only male) BG control, n = 90 BG fluctuation, n = 120	210	BG controls: 55.53 ± 7.30 BG fluctuation: 56.41 ± 7.67	BG controls: 6.59 ± 2.30 BG fluctuation group: 6.92 ± 2.25	nr	2 days
He, 2023	Longitudinal (prospective) FU: 1 year	T2D with kidney disease on hemodialysis High TIR, n = 12 Low TIR, n = 15	27	High TIR: 66 [63, 73] Low TIR: 70 [64, 75]	High TIR: 2 [1.7, 10] Low TIR: 5 [0.75, 20]	High TIR: 43 [37, 51] mmol/mol (6.1 [5.5, 6.8]%) Low TIR: 66 [46, 70] mmol/mol (8.2 [6.4, 8.6]%)	14 days
Lu, 2021	Longitudinal (prospective) FU: until death occurred or 3–13 years Median: 6.9 years	T2D, hospitalized	6225	mean age = 61.7 years	mean: 9.7 years	74.0 ± 24.0 mmol/mol (8.9 ± 2.2%)	72 hours
Wei, 2019	Longitudinal (prospective) Median FU [IQR]: 31 months (22, 56)	T2D Divided into three groups: 1) No hypo, n = 1173 2) Mild hypo (lvl1), n = 323 3) Severe hypo (lvl3), n = 24	1520	No hypoglycemia: 58.59 ± 11.26 Hypoglycemia: 62.27 ± 11.58	No hypoglycemia: 6.46 ± 6.00 Hypoglycemia: 7.78 ± 7.37	No hypoglycemia: (8.19 ± 2.10%) Hypoglycemia: (7.73 ± 1.96%)	3 days

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Bezerra, 2023	Cross-sectional	T1D	161	37.4 ± 13.4	17.7 ± 10.6	$(7.5 \pm 1.1\%)$	14 days
De Meule- meester, 2024 [†]	Cross-sectional	T1D	808	44.8 ± 15.2	23.1 ± 13.6	63 ± 13 mmol/mol $(7.9 \pm 1.2\%)$	2 weeks
Deng, 2023	Cross-sectional	T2D	860	HP 1 carriers: 53.5 ± 13.3 HP 2-2: 51.7 ± 14.6	HP 1 carriers: 8.6 ± 6.4 HP 2-2: 8.6 ± 6.7	HP1 carriers: 73.0 ± 24.0 mmol/mol $(8.8 \pm 2.2\%)$ Hp2-2: 72.0 ± 23.0 mmol/mol (8.7 ± 2.1)	3 days
El Malahi, 2022	Cross-sectional	T1D starting on sensor-augmented pump therapy	515	Mean $\pm$ SD [years]: 42.2 ± 12.5	22.3 ± 11.6	60 ± 9.8 mmol/mol $(7.6 \pm 0.9 \%)$	2 weeks
Guo, 2021	Cross-sectional	T1D or T2D with atrial fibrillation With Stroke, n = 48 Without Stroke, n = 462	510	Stroke: 70.3 ± 12.1 No stroke: 68.1 ± 9.4	nr	No Stroke: 7.4 ± 2.1 Stroke: 8.2 ± 1.7	72 hour
Li, 2020	Cross-sectional	T2D with LEAD, n = 179 T2D without LEAD, n = 157	336	Without LEAD: 55.94 ± 12.45 With LEAD: 65.56 ± 11.99	Without LEAD: 6.92 ± 3.54 With LEAD: 10.32 ± 4.14	Without LEAD: $7.85 \pm 1.41\%$ With LEAD: $8.97 \pm 1.63\%$	72 hours

Table 4.3: Study design and demographics of the included clinical cardiovascular outcome studies

Author, year	Study Design	Population	Size	Age (Years)	Diabetes duration (Years)	HbA1c, mmol/mol (%)	CGM duration
Magri, 2018 [†]	Cross-sectional	T2D	121	64 [57, 68]	3 [2, 5]	median: 45 mmol/mol (6.8%)	72 hours
Mi, 2012	Cross-sectional	T2D with chest pain Without CAD, n = 202 With CAD, n = 84	286	Without CAD: 62.8 ± 8.7 With CAD: 66.6 ± 9.2	nr	Non-CAD: 7.51 ± 0.80% CAD: 7.75 ± 0.92%	72 hours (Only intermediate 48 hours were used)
Sheng, 2023	Cross-sectional	T2D, hospitalized	545	mean age of (61.22 ± 11.21) years	nr	8.51 ± 1.85%	7-14 days
Su, 2011	Cross-sectional	T2D with chest pain Without CAD, n = 92 With CAD, n = 252	344	Without CAD: 61 ± 9 With CAD: 65 ± 9	Without CAD: 4.8 ± 5.7 With CAD: 6.5 ± 6.4	non-CAD: 7.5 ± 1.4% CAD: 7.6 ± 1.5%	72 hours (only 48 hours used)
Watanabe, 2017	Cross-sectional	Prediabetes, hospitalized	28	64.3 ± 12.8	nr	5.41 ± 0.35%	72 hours (Only used the middle 48 hours)

Zhang, 2013	Cross-sectional	T2D with cardiovascular complications Group A: healthy individuals Group B: T2D without cardiovascular complications Group C: T2D with cardiovascular complications	92	Group A: 56.3 ± 6.1 Group B: 56.1 ± 6.6 Group C: 61.7 ± 7.2	nr	Group A: 5.3 ± 0.3% Group B: 6.6 ± 1.2% Group C: 7.5 ± 1.4%	72 hours
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[†] This study appears both in the clinical and subclinical disease outcome tables due to investigating multiple CVD outcomes. Note: Age is reported in years either as an interval, mean ± SD, or median (IQR). Diabetes duration values originally reported in months were converted to years for consistency (months ÷ 12).
Abbreviations: CGM: continuous glucose monitoring, FU: follow-up, T1D: type 1 diabetes, T2D: type 2 diabetes, CAD: coronary artery disease, hypo: hypoglycemia, LEAD: lower extremity arterial disease, IQR: interquartile range, nr: not reported.

Clinical cardiovascular outcomes comprised cardiovascular mortality, major adverse cardiovascular events, macrovascular complications, coronary artery disease, stroke, and peripheral artery disease. Most studies were observational and differed substantially in study populations, CGM-derived metrics, outcome definitions, and adjustment strategies. Consequently, direct comparison across studies was challenging. The reported effect estimates from the least and most adjusted analyses are summarized in Table 4.4.

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
Cardiovascular Mortality					
Lu et al ²¹ , 2021					Model 1: Demographic Model 2: Demographic
TIR > 85%	HR 1.00	< 0.001 (trend)	HR 1.00	0.02 (trend)	Anthropometric
TIR 71% – 85%	HR: 1.43 (0.95–2.14)	< 0.001 (trend)	HR: 1.35 (0.90–2.04)	0.02 (trend)	Lifestyle
TIR 51% – 70%	HR: 1.66 (1.12–2.45)	< 0.001 (trend)	HR: 1.47 (0.99–2.19)	0.02 (trend)	Medical History
TIR ≥ 50%	HR: 2.15 (1.47–3.13)	< 0.001 (trend)	HR: 1.85 (1.25–2.72)	0.02 (trend)	Physiological
TIR as a continuous variable (each 10% decrease)	HR 1.08 (1.03–1.13)	—	HR 1.05 (1.00–1.11)	—	Biochemical Medication
Wei et al ^{Wei2019?*} , 2019					Least adjusted: Unadjusted Most adjusted: Demographic
Hypoglycemic events	HR: 2.03 (1.21–3.41)	—	HR: 2.64 (1.40–4.99)	—	Anthropometric Lifestyle Medical History Biochemical Medication
Major Adverse Cardiovascular Events					
He et al ^{He2023?} , 2023					
Blood glucose risk index	HR: 0.97 (0.85–1.10)	0.61	HR: 0.98 (0.85–1.13)	0.75	
Low blood glucose index	HR: 2.37 (1.16–4.83)	0.02	HR: 2.73 (1.21–6.16)	0.02	
High blood glucose index	HR: 0.94 (0.81–1.08)	0.38	HR: 0.94 (0.81–1.09)	0.44	
Average of daily risk range	HR: 1.00 (0.93–1.07)	> 0.99	HR: 1.01 (0.93–1.09)	0.80	

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
GMI	HR: 0.98 (0.91–1.06)	0.65	HR: 0.99(0.91–1.07)	0.78	
M-value	HR: 0.98 (0.91–1.05)	0.54	HR: 0.98(0.91–1.06)	0.64	
Wei et al ^{Wei2019?*, 2019}					
Hypoglycemic events	HR: 1.501 (1.207–1.866)	—	HR: 1.615 (1.239–2.106)	< 0.001	
Macrovascular Complications					
De Meulemeester et al ^{Meulemeester2024?†,*, 2024}					
TIR	OR: 0.939 (0.829–1.063)	> 0.05	OR: 0.896 (0.738–1.087)	> 0.05	
TITR	OR: 0.901 (0.775–1.047)	> 0.05	OR: 0.933 (0.745–1.169)	> 0.05	
Bezerra et al ^{Bezerra2023?, 2023}					
TIR	OR: 0.66 (0.46–0.93)	0.02	OR: 0.68 (0.39–1.16)	0.15	
Time below 54 mg/dL	OR: 1.10 (0.88–1.38)	0.39	OR: 0.92 (0.62–1.34)	0.65	
TBR	OR: 0.93 (0.80–1.09)	0.39	OR: 0.77 (0.54–1.11)	0.17	
TAR	OR: 1.04 (1.01–1.08)	0.01	OR: 1.04 (0.99–1.10)	0.08	
Time above 250 mg/dL	OR: 1.04 (1.00–1.08)	0.03	OR: 1.03 (0.97–1.09)	0.29	
CV	OR: 1.03 (0.95–1.11)	0.52	OR: 0.92 (0.81–1.06)	0.25	
GMI	OR: 2.17 (1.14–4.11)	0.02	OR: 2.03 (0.77–5.37)	0.15	
Deng et al ^{Deng2023?, 2023}					
% CV tertile 1 (Hp1; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	
% CV tertile 1 (Hp2-2; reference)	OR: 1.000	0.07 (interaction)	OR: 1.000	0.008 (interaction)	
% CV tertile 2 (Hp1)	OR: 1.483 (0.907–2.423)	0.12	OR: 1.048 (0.528–2.078)	0.89	
% CV tertile 2 (Hp2-2)	OR: 1.399 (0.829–2.358)	0.21	OR: 0.659 (0.296–1.466)	0.31	

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
% CV tertile 3 (Hp1)	OR: 2.347 (1.393–3.957)	0.001	OR: 2.461 (1.183–5.121)	0.02	
% CV tertile 3 (Hp2-2)	OR: 1.217 (0.731–2.027)	0.45	OR: 0.540 (0.245–1.191)	0.13	
El Malahi et al ^{Malahi2022?} , 2022					
TIR	—	—	—	> 0.05	
SD	—	—	—	> 0.05	
CV	—	—	—	> 0.05	
Magri et al ^{27†} , 2018					
TBR	—	—	OR: 1.12 (1.014–1.228)	0.02	
Lowest BG value	—	—	—	—	
Area under the TBR curve	—	—	—	—	
Coronary Artery Disease					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TITR	OR: 1.039 (0.812–1.330)	> 0.05	OR: 1.255 (0.874–1.803)	> 0.05	
TIR	OR: 1.072 (0.866–1.328)	> 0.05	OR: 1.164 (0.844–1.607)	> 0.05	
Sheng et al ^{Sheng2023?} , 2023					
TIR < 20%	—	—	OR: 2.143 (1.554–3.287)	—	
TIR 20% – 40%	—	—	OR: 1.049 (0.945–2.022)	—	
TIR 40% – 60%	—	—	OR 0.854 (0.495–1.473)	—	

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
TIR 60% – 80%	—	—	OR: 0.617 (0.423–1.312)	—	
TIR > 80%	—	—	OR: 0.470 (0.143–1.545)	—	
Chen et al ^{Chen2020?^{†,*}, 2020}					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 5.797$	0.02	
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L versus high BG fluctuations (angina pectoris)	—	—	$\chi^2 = 7.490$	0.006	
Wei et al ^{Wei2019?[*], 2019}					
Hypoglycemic events (myocardial Infarction)	HR: 1.901 (1.067–3.389)	—	HR: 1.549 (0.768–3.124)	0.03	
Hypoglycemic events (unstable angina pectoris)	HR: 1.226 (0.857–1.753)	—	HR: 1.218 (0.794–1.869)	0.30	
Shu-Hua et al ^{Hua?^e, 2012}					
MAGE level (≥ 3.4 mmol/L)	—	—	OR: 2.286 (1.176–4.446)	0.02	
Su et al ^{28[*], 2011}					

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
MAGE ≥ 3.4 mmol/L	—	—	OR: 2.612 (1.423–4.831)	0.002	
MAGE	—	—	AUC: 0.618 (0.555–0.680)	0.001	
Gensini Score					
Chen et al ^{Chen2020?†,*} , 2020					
Controls with SD < 1.40 mmol/L, MAGE < 3.90 mmol/L, LAGE < 4.40 mmol/L, MODD < 0.83 mmol/L	—	—	$t = 6.210$	< 0.001	
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	$r = 0.742$	< 0.001	
Shu-Hua et al ^{Hua2012?*} , 2012					
MAGE	—	—	Unstandardized coefficient $\beta = 4.817$; SE = 1.164; Standardized coefficient $t = 2.984$	0.003	
Su et al ^{28*} , 2011					
MAGE	—	—	Unstandardized $\beta = 7.010$; SE = 1.466; standardized $\beta = 0.237$;	< 0.001	

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
<i>t</i> = 4.783					
Syntax Score					
Watanabe et al ^{Watanabe2017?*} , 2017					
MAGE	—	—	<i>r</i> = 0.776	< 0.001	
Zhang et al ^{30*} , 2013					
MAGE	—	—	<i>r</i> = 0.518	0.01	
BG fluctuations from 00:00 to 03:00	—	—	<i>r</i> = −0.442	0.04	
BG fluctuations from 03:00 to 06:00	—	—	<i>r</i> = −0.208	0.34	
BG fluctuations from 06:00 to 08:00	—	—	<i>r</i> = 0.678	< 0.001	
BG fluctuations from 08:00 to 11:00	—	—	<i>r</i> = 0.115	0.60	
BG fluctuations from 11:00 to 13:00	—	—	<i>r</i> = 0.523	0.01	
BG fluctuations from 13:00 to 17:00	—	—	<i>r</i> = 0.257	0.24	
BG fluctuations from 17:00 to 19:00	—	—	<i>r</i> = 0.358	0.09	
BG fluctuations from 19:00 to 24:00	—	—	<i>r</i> = −0.018	0.93	
Stroke					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
TITR	OR 0.651 (0.470–0.902)	< 0.05	OR 0.546 (0.347–0.858)	< 0.01	
TIR	OR 0.749 (0.588–0.955)	< 0.05	OR 0.617 (0.440–0.866)	< 0.01	
Guo et al ^{Guo2021?*} , 2021					
TIR: Q1 ($\leq 46\%$; reference)	OR: 1.00	< 0.001	OR: 1.00	< 0.001	
TIR: Q2 ($46\% - 65\%$)	OR: 0.86 (0.72–0.95)	< 0.001	OR: 0.80 (0.68–0.92)	< 0.001	
TIR: Q3 ($65\% - 81\%$)	OR: 0.71 (0.61–0.81)	< 0.001	OR: 0.64 (0.53–0.79)	< 0.001	
TIR: Q4 ($> 81\%$)	OR: 0.66 (0.58–0.80)	< 0.001	OR: 0.59 (0.50–0.74)	< 0.001	
TIR (per 10% increase)	OR: 0.93 (0.85–0.98)	0.008	OR: 0.89 (0.82–0.95)	< 0.001	
Wei et al ^{Wei2019?*} , 2019					
Hypoglycemic events	HR: 1.691 (1.144–2.499)	—	HR: 1.813 (1.110–2.960)	0.06	
Peripheral artery disease					
De Meulemeester et al ^{Meulemeester2024?†,*} , 2024					
TITR	OR 0.680 (0.426–1.085)	> 0.05	OR 0.807 (0.382–1.703)	> 0.05	
TIR	OR 0.736 (0.520–1.042)	> 0.05	OR 0.811 (0.470–1.398)	> 0.05	
Lower Extremity Arterial Disease					
Li et al ^{Li2020?} , 2020					
TIR	OR: 0.979 (0.968–0.991)	< 0.001	OR: 0.979 (0.965–0.992)	0.002	
CV	OR: 1.040 (1.003–1.078)	0.04	OR: 1.038 (0.996–1.081)	0.08	

Table 4.4: Main findings of the included studies on clinical cardiovascular outcomes. WIP: Still need to add suppl. files, and adjustment column

Outcome, reference, and continuous glucose monitoring metrics	Unadjusted or least adjusted findings	<i>P</i> value for the least adjusted findings	Most adjusted findings	<i>P</i> value for the most adjusted findings	Adjustments
SD	OR: 1.325 (1.038–1.691)	0.02	OR: 1.158 (0.824–1.627)	0.40	
TIR-without LEAD (1)	OR: 1.00	—	OR: 1.00	—	
TIR-mild LEAD (1)	OR: 0.98 (0.97–1.00)	0.14	OR: 0.99 (0.97–1.01)	0.25	
TIR-moderate LEAD (1)	OR: 0.97 (0.95–0.99)	0.007	OR: 0.97 (0.95–0.99)	0.01	
TIR-without severe LEAD (1)	OR: 0.96 (0.94–0.98)	0.002	OR: 0.96 (0.94–0.98)	0.003	
CV-without LEAD	—	—	OR: 1.00	—	
CV-mild LEAD	—	—	OR: 1.03 (0.98–1.07)	0.28	
CV-moderate LEAD	—	—	OR: 1.02 (0.96–1.09)	0.48	
CV-without severe LEAD	—	—	OR: 1.02 (0.95–1.09)	0.60	
TIR-without LEAD (2)	—	—	OR: 1.00	—	
TIR-mild LEAD (2)	—	—	OR: 0.97 (0.96–1.08)	0.06	
TIR-moderate LEAD (2)	—	—	OR: 0.98 (0.95–0.99)	0.01	
TIR-without severe LEAD (2)	—	—	OR: 0.97 (0.95–0.99)	0.02	
SD-without LEAD	—	—	OR: 1.00	—	
SD-mild LEAD	—	—	OR: 0.88 (0.47–1.64)	0.69	
SD-moderate LEAD	—	—	OR: 1.28 (0.58–3.07)	0.58	
SD-without severe LEAD	—	—	OR: 1.52 (0.92–2.41)	0.10	

All hazard ratios (HRs), odds ratios (ORs), and areas under the curve (AUCs) are reported as follows: point estimate (95% CI).

* This study appears multiple times as it investigated multiple cardiovascular disease outcomes.

† This study appears in both the clinical and subclinical disease outcome tables owing to the investigation of multiple cardiovascular disease outcomes.

The full table with adjustments is provided in **Multimedia Appendix 10**. Further elaboration on the adjusted variables can be found in **Multimedia Appendices 7 and 8**.

*Abbreviations: * nr: Not applicable or not available/not reported, TIR: time in range, GMI: glucose management indicator, TITR: time in tight range, TBR: time below range, TAR: time above range, CV: coefficient of variation, BG: blood glucose, MAGE: mean amplitude of glycemic excursions, LAGE: largest amplitude of glycemic excursions, MODD: mean of daily differences, LEAD: lower extremity arterial disease.

4.2 Baseline Characteristics Tables of the Datasets

The CANCAN dataset presented below was used in Study II (Table 4.12). The total number of included participants were 159 and 67 (42%) had elevated NT-proBNP values. Participants with elevated NT-proBNP tended to be older, have a longer duration of diabetes, and more frequently have cardiovascular disease. They also tended to have lower BMI and lower LDL and total cholesterol concentrations, while exhibiting modestly better overall glycaemic control but slightly higher glycaemic variability.

Table 4.5: Baseline table for CANCAN data.

Variable	Total	NT-proBNP \$ \$125 pg/mL	NT-proBNP >125 pg/mL
N	159	92	67
Age (years)	65 [56, 71]	60 [53, 68]	68 [64, 76]
Sex			
– Male	99 (62%)	55 (60%)	44 (66%)
– Female	60 (38%)	37 (40%)	23 (34%)
Smoking status			
– Never or former	135 (85%)	77 (84%)	58 (87%)
– Current	24 (15%)	15 (16%)	9 (13%)
Diabetes duration (years)	16 [11, 24]	14 [9, 22]	19 [14, 25]
SCORE2-Diabetes (%)	14.8 [9.1, 20.9]	11.9 [7.8, 17.2]	19.0 [13.4, 25.4]
HbA1c (mmol/mol)	64 [55, 80]	66 [56, 86]	60 [54, 72]
Systolic blood pressure (mmHg)	132 [123, 143]	132 [123, 143]	131 [122, 142]
HDL (mmol/L)	1.0 [0.9, 1.2]	1.0 [0.9, 1.2]	1.0 [0.9, 1.3]
Total cholesterol (mmol/L)	3.9 [3.2, 4.8]	4.1 [3.5, 4.9]	3.6 [3.0, 4.2]
eGFR (mL/min/1.73 m ²)	57 [39, 74]	68 [52, 82]	48 [27, 63]
CGM-derived metrics			
– CGM days (n/participant)	13 [10, 13]	13 [9, 13]	13 [10, 13]
– Glucose management indicator (%)	7.3 [6.8, 8.4]	7.5 [6.8, 8.6]	7.2 [6.7, 8.0]
– Time in range (%)	59.6 [31.9, 78.9]	58.2 [20.2, 78.6]	62.1 [40.8, 81.0]
– Time below range (%)	0.0 [0.0, 1.5]	0.0 [0.0, 0.8]	0.2 [0.0, 2.0]
– Time above range (%)	36.6 [17.6, 68.1]	41.8 [19.8, 79.5]	33.1 [14.1, 58.2]
– Coefficient of variation (%)	27.3 [22.5, 31.9]	25.3 [21.2, 30.3]	28.3 [24.8, 34.2]
– Mean amplitude of glycemic excursions (mmol/L)	6.3 [5.0, 7.9]	6.0 [4.9, 7.7]	6.7 [5.1, 8.5]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

The T1DX dataset presented below was used in Study II and Study III (Table 4.13). A total of 205 participants were included, of whom 101 (49%) were White. Compared with Black participants, White participants tended to have lower HbA1c levels (63 mmol/mol vs. 72 mmol/mol) and a higher proportion of children (49% vs. 32%). Females constituted the majority of both racial groups (46% and 40%).

Table 4.6: Baseline table for T1DX data.

Characteristics	Total	Race: White	Race: Black
N	205	101	104

Table 4.6: Baseline table for T1DX data.

Characteristics	Total	Race: White	Race: Black
Race (White)	101 (49%)	101 (100%)	0 (0%)
Sex (male)	88 (43%)	46 (46%)	42 (40%)
Age (years)	28 [15, 47]	22 [13, 46]	32 [16, 47]
Age group (child, < 18 yrs)	82 (40%)	49 (49%)	33 (32%)
Age at T1D diagnosis (years)	11 [7, 22]	9 [6, 16]	14 [8, 25]
BMI (kg/m ²)	25.1 [21.7, 29.2]	24.0 [20.7, 28.6]	26.7 [22.3, 30.0]
HbA1c (mmol/mol)	67 [57, 78]	63 [55, 73]	72 [60, 85]
CGM-derived metrics			
– CGM days (n/participant)	58 [46, 69]	63 [47, 71]	56 [45, 66]
– Glucose management indicator (%)	7.5 [7.0, 8.2]	7.4 [6.9, 7.9]	7.7 [7.0, 8.5]
– Time in range (%)	47 [35, 57]	49 [39, 60]	42 [31, 54]
– Time above range (%)	45 [30, 56]	42 [28, 53]	48 [32, 60]
– Time below range (%)	8 [4, 14]	6 [4, 10]	11 [5, 15]
– Coefficient of variation (%)	43.3 [39.6, 48.6]	42.2 [38.6, 46.5]	46.7 [40.4, 51.9]
– Mean amplitude of glycemic excursions (mmol/L)	9.9 [8.2, 12.1]	9.6 [8.1, 11.2]	10.6 [8.5, 13.0]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

4.3 Study II: Results from a Foundation Model Embedding Analysis

4.4 Study II: Prediction of NT-proBNP

Results from the prediction analysis are shown in Table 4.14. The highest optimism-corrected AUROC was achieved by Model A2 which included both all nine clinical variables from the SCORE2-Diabetes input and the CGM-derived metrics TIR and MAGE (AUROC: 0.81 [0.77–0.88]). Optimism-corrected AUROCs for the models with fewer predictors had modest performance (AUROC: 0.72 to 77), however the optimism-correction was smaller and the prediction stability was more stable the fewer predictors were included in the models (Models: B1, B2, and B3) (@#sec-suppl-pred-stability). The models including only CGM-data either as derived metrics (Model C1) or as UMAP dimensions (Model D1a, D1b, D1c) performed the worst (AUROC: 0.53 to 0.69). Age seemed a better predictor for elevated NT-proBNP (AUROC: 0.73 [0.65 - 0.80])

Table 4.7: Area under the receiver operating characteristic curve for the included models.

Models	Corrected [95% CI]	Apparent	Optimism	Number of predictors
Model A0: (Age)	0.73 [0.65 - 0.80]	0.73	0.00	1
Model A1: (SCORE2-Diabetes input)	0.79 [0.76 - 0.87]	0.83	0.04	9
Model A2: (Model A1 + TIR and MAGE)	0.81 [0.77-0.88]	0.86	0.05	11
Model A3a: (Model A1 + UMAP (min))	0.80 [0.75 - 0.87]	0.84	0.05	11
Model A3b: (Model A1 + UMAP (max))	0.79 [0.75 - 0.87]	0.84	0.05	11

Table 4.7: Area under the receiver operating characteristic curve for the included models.

Models	Corrected [95% CI]	Apparent	Optimism	Number of predictors
Model A3c: (Model A1 + UMAP (mean))	0.79 [0.75 - 0.87]	0.84	0.05	11
Model B1: (SCORE2-Diabetes score)	0.74 [0.65 - 0.81]	0.73	0.00	1
Model B2: (Model B1 + TIR and MAGE)	0.77 [0.70 - 0.85]	0.79	0.01	3
Model B3a: (Model B1 + UMAP (min))	0.72 [0.65 - 0.80]	0.74	0.02	3
Model B3b: (Model B1 + UMAP (max))	0.72 [0.65 - 0.81]	0.74	0.02	3
Model B3c: (Model B1 + UMAP (mean))	0.73 [0.65 - 0.81]	0.75	0.02	3
Model C1: (TIR and MAGE)	0.69 [0.61 - 0.77]	0.69	0.01	11
Model D1a: (UMAP, max*)	0.55 [0.48 - 0.64]	0.58	0.03	2
Model D1b: (UMAP, min*)	0.53 [0.43 - 0.63]	0.57	0.04	2
Model D1c: (UMAP, mean*)	0.53 [0.43 - 0.64]	0.56	0.03	2

* Indicates the pooling technique for the embeddings. * Abbreviations: * CI: Confidence Interval, CGM: Continuous Glucose Monitoring, UMAP: Uniform Manifold Approximation and Projection.

4.5 Study III: Results from an Algorithmic Fairness Study

Bias is an important spect to consider when working data drivenly. In study III, we therefore evaluates bias in CGM trained machine learning models for forecasting in the T1DX cohort

4.5.1 Bias in Machine Learning Models

The RMSE of the models are shown in Table 4.15 and Table 4.16. Base-Generalized and Transfer Learned models achieved lower prediction errors (RMSE: 2.00 to 2.06 mmol/L) than the naive last observation carried forward (LOCF)(RMSE: 2.41 and 2.45 mmol/L) and Base-Individual models (RMSE: 2.54 to 2.63 mmol/L) for both White and Black participants (Table 4.15 and Table 4.16).

Table 4.8: Root mean squared errors with 95% CIs for the different models evaluated on White participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
10			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
20			2.04 [1.95, 2.12]	2.02 [1.94, 2.10]
30			2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
40	2.45 [2.32, 2.57]	2.54 [2.38, 2.70]	2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
50	(independent	(independent	2.03 [1.94, 2.11]	2.02 [1.93, 2.10]
60	of ratio)	of ratio)	2.02 [1.94, 2.11]	2.02 [1.93, 2.10]

Table 4.8: Root mean squared errors with 95% CIs for the different models evaluated on White participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
70			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
80			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
90			2.01 [1.92, 2.10]	2.01 [1.93, 2.10]
100			2.01 [1.92, 2.10]	2.01 [1.92, 2.10]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40

Table 4.9: Root mean squared errors with 95% CIs for the different models evaluated on Black participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
10			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
20			2.05 [1.96, 2.14]	2.00 [1.92, 2.08]
30			2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
40	2.41 [2.28, 2.53]	2.63 [2.47, 2.79]	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
50	(independent	(independent	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
60	of ratio)	of ratio)	2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
70			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
80			2.06 [1.97, 2.14]	2.00 [1.92, 2.09]
90			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]
100			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40

For the Base-Generalized models, prediction performance gradually diverged between White and Black participants as the proportion of White participants in the training data increased (Figure 4.4). For example, when the training data changed from 0% to 100% White participants, the difference in RMSE between the racial groups increased by 0.04 mmol/L [95% CI: 0.001 to 0.008], $p=0.02$, indicating that training data composition affected the two groups differently (Table 4.17). This divergence was not observed after transfer learning (0.002 mmol/L [95% CI: -0.001 to 0.005], $p=0.20$) (Table 4.18). A statistically significant differences were however found in both the Base-Generalized and the Transfer Learned models for age groups (0.482 [0.355 to 0.610], $p < 0.001$, Table 4.17 and 0.411 [0.290 to 0.533], $p < 0.001$, Table 4.18 respectively). The differences in age groups showed a lower error when forecasting blood glucose values for adults compared to children. Larger sample sizes available for transfer learning were associated with lower prediction error ($\beta = -0.111$ mmol/L [-0.161 to -0.061] mmol/L, $p < 0.001$) for every additional 1,000 samples used for fine-tuning (Table 4.18). In contrast, the size of the dataset used to train the Base-Generalized model was not associated with prediction error.

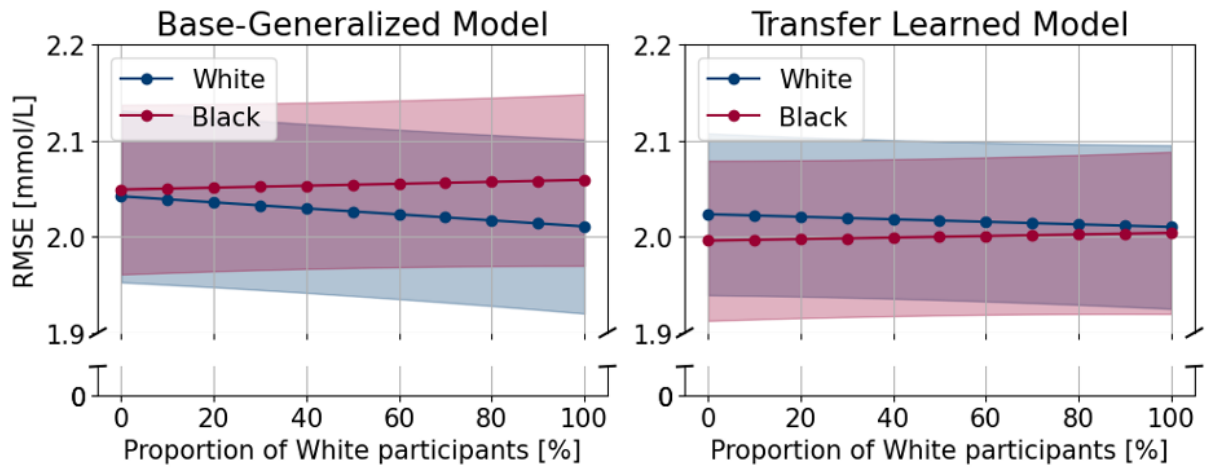


Figure 4.3: Estimated root mean squared errors (RMSE) across 11 different proportions of White and Black participants in the dataset.

Table 4.10: Model coefficients from linear mixed effects models for root mean squared errors of the Base-Generalized models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.893	[1.762, 2.024]	<0.001
Race			
White	Reference		
Black	0.007	[-0.118, 0.132]	0.91
Age			
Adult	Reference		
Child	0.482	[0.355, 0.610]	<0.001
Sex			
Male	Reference		
Female	-0.076	[-0.201, 0.048]	0.23
Training size Base-Generalized*	0.019	[-0.055, 0.093]	0.62
Ratio (per 10%)	-0.003	[-0.007, 0.001]	0.12
Ratio · Race	0.004	[0.001, 0.008]	0.02

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

Table 4.11: Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.890	[1.768, 2.015]	<0.001
Race			
White	Reference		
Black	-0.028	[-0.147, 0.092]	0.65
Age			
Adult	Reference		
Child	0.411	[0.290, 0.533]	<0.001
Sex			
Male	Reference		

Table 4.11: Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

Predictor	Coefficient	95% CI	P-value
Female	-0.057	[-0.173, 0.060]	0.34
Training size Base-Generalized*	0.011	[-0.060, 0.081]	0.77
Training size tl**	-0.111	[-0.161, -0.061]	<0.001
Ratio (per 10 units)	-0.001	[-0.003, 0.001]	0.46
Ratio · Race	0.002	[-0.001, 0.005]	0.20

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

** Per 1,000 samples. (Training size for the transfer learned approach was centered at 4,400 as the training data for fine-tuning had a mean of 4,371 samples.)

4.5.2 The Surveillance Error Grid

Surveillance Error Grid analysis showed minimal differences between models and training data compositions. Across all models, racial groups, and training data ratios, almost all predictions fell within the “None” or “Slight” risk zones, while approximately 3–6% were classified as “Moderate” risk and no predictions were classified as “High” or “Extreme” risk (Appendix F). These findings indicate that the small differences in RMSE observed between models were unlikely to be clinically meaningful.

4.6 Baseline Characteristics Tables of the Datasets

The CANCAN dataset presented below was used in Study II (Table 4.12). The total number of included participants were 159 and 67 (42%) had elevated NT-proBNP values. Participants with elevated NT-proBNP tended to be older, have a longer duration of diabetes, and more frequently have cardiovascular disease. They also tended to have lower BMI and lower LDL and total cholesterol concentrations, while exhibiting modestly better overall glycaemic control but slightly higher glycaemic variability.

Table 4.12: Baseline table for CANCAN data.

Variable	Total	NT-proBNP \$ \$125 pg/mL	NT-proBNP >125 pg/mL
N	159	92	67
Age (years)	65 [56, 71]	60 [53, 68]	68 [64, 76]
Sex			
– Male	99 (62%)	55 (60%)	44 (66%)
– Female	60 (38%)	37 (40%)	23 (34%)
Smoking status			
– Never or former	135 (85%)	77 (84%)	58 (87%)
– Current	24 (15%)	15 (16%)	9 (13%)
Diabetes duration (years)	16 [11, 24]	14 [9, 22]	19 [14, 25]
SCORE2-Diabetes (%)	14.8 [9.1, 20.9]	11.9 [7.8, 17.2]	19.0 [13.4, 25.4]
HbA1c (mmol/mol)	64 [55, 80]	66 [56, 86]	60 [54, 72]
Systolic blood pressure (mmHg)	132 [123, 143]	132 [123, 143]	131 [122, 142]
HDL (mmol/L)	1.0 [0.9, 1.2]	1.0 [0.9, 1.2]	1.0 [0.9, 1.3]
Total cholesterol (mmol/L)	3.9 [3.2, 4.8]	4.1 [3.5, 4.9]	3.6 [3.0, 4.2]

Table 4.12: Baseline table for CANCAN data.

Variable	Total	NT-proBNP \$ \$125 pg/mL	NT-proBNP >125 pg/mL
eGFR (mL/min/1.73 m ²)	57 [39, 74]	68 [52, 82]	48 [27, 63]
CGM-derived metrics			
– CGM days (n/participant)	13 [10, 13]	13 [9, 13]	13 [10, 13]
– Glucose management indicator (%)	7.3 [6.8, 8.4]	7.5 [6.8, 8.6]	7.2 [6.7, 8.0]
– Time in range (%)	59.6 [31.9, 78.9]	58.2 [20.2, 78.6]	62.1 [40.8, 81.0]
– Time below range (%)	0.0 [0.0, 1.5]	0.0 [0.0, 0.8]	0.2 [0.0, 2.0]
– Time above range (%)	36.6 [17.6, 68.1]	41.8 [19.8, 79.5]	33.1 [14.1, 58.2]
– Coefficient of variation (%)	27.3 [22.5, 31.9]	25.3 [21.2, 30.3]	28.3 [24.8, 34.2]
– Mean amplitude of glycemic excursions (mmol/L)	6.3 [5.0, 7.9]	6.0 [4.9, 7.7]	6.7 [5.1, 8.5]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

The T1DX dataset presented below was used in Study II and Study III (Table 4.13). A total of 205 participants were included, of whom 101 (49%) were White. Compared with Black participants, White participants tended to have lower HbA1c levels (63 mmol/mol vs. 72 mmol/mol) and a higher proportion of children (49% vs. 32%). Females constituted the majority of both racial groups (46% and 40%).

Table 4.13: Baseline table for T1DX data.

Characteristics	Total	Race: White	Race: Black
N	205	101	104
Race (White)	101 (49%)	101 (100%)	0 (0%)
Sex (male)	88 (43%)	46 (46%)	42 (40%)
Age (years)	28 [15, 47]	22 [13, 46]	32 [16, 47]
Age group (child, < 18 yrs)	82 (40%)	49 (49%)	33 (32%)
Age at T1D diagnosis (years)	11 [7, 22]	9 [6, 16]	14 [8, 25]
BMI (kg/m ²)	25.1 [21.7, 29.2]	24.0 [20.7, 28.6]	26.7 [22.3, 30.0]
HbA1c (mmol/mol)	67 [57, 78]	63 [55, 73]	72 [60, 85]
CGM-derived metrics			
– CGM days (n/participant)	58 [46, 69]	63 [47, 71]	56 [45, 66]
– Glucose management indicator (%)	7.5 [7.0, 8.2]	7.4 [6.9, 7.9]	7.7 [7.0, 8.5]
– Time in range (%)	47 [35, 57]	49 [39, 60]	42 [31, 54]
– Time above range (%)	45 [30, 56]	42 [28, 53]	48 [32, 60]
– Time below range (%)	8 [4, 14]	6 [4, 10]	11 [5, 15]
– Coefficient of variation (%)	43.3 [39.6, 48.6]	42.2 [38.6, 46.5]	46.7 [40.4, 51.9]
– Mean amplitude of glycemic excursions (mmol/L)	9.9 [8.2, 12.1]	9.6 [8.1, 11.2]	10.6 [8.5, 13.0]

Categorical values are given by number (percentage), and numerical values are given by median [interquartile range].

4.7 Study II: Results from a Foundation Model Embedding Analysis

4.8 Study II: Prediction of NT-proBNP

Results from the prediction analysis are shown in Table 4.14. The highest optimism-corrected AUROC was achieved by Model A2 which included both all nine clinical variables from the SCORE2-Diabetes input and the CGM-derived metrics TIR and MAGE (AUROC: 0.81 [0.77–0.88]). Optimism-corrected AUROCs for the models with fewer predictors had modest performance (AUROC: 0.72 to 0.77), however the optimism-correction was smaller and the prediction stability was more stable the fewer predictors were included in the models (Models: B1, B2, and B3) (@#sec-suppl-pred-stability). The models including only CGM-data either as derived metrics (Model C1) or as UMAP dimensions (Model D1a, D1b, D1c) performed the worst (AUROC: 0.53 to 0.69). Age seemed a better predictor for elevated NT-proBNP (AUROC: 0.73 [0.65 - 0.80])

Table 4.14: Area under the receiver operating characteristic curve for the included models.

Models	Corrected [95% CI]	Apparent	Optimism	Number of predictors
Model A0: (Age)	0.73 [0.65 - 0.80]	0.73	0.00	1
Model A1: (SCORE2-Diabetes input)	0.79 [0.76 - 0.87]	0.83	0.04	9
Model A2: (Model A1 + TIR and MAGE)	0.81 [0.77-0.88]	0.86	0.05	11
Model A3a: (Model A1 + UMAP (min))	0.80 [0.75 - 0.87]	0.84	0.05	11
Model A3b: (Model A1 + UMAP (max))	0.79 [0.75 - 0.87]	0.84	0.05	11
Model A3c: (Model A1 + UMAP (mean))	0.79 [0.75 - 0.87]	0.84	0.05	11
Model B1: (SCORE2-Diabetes score)	0.74 [0.65 - 0.81]	0.73	0.00	1
Model B2: (Model B1 + TIR and MAGE)	0.77 [0.70 - 0.85]	0.79	0.01	3
Model B3a: (Model B1 + UMAP (min))	0.72 [0.65 - 0.80]	0.74	0.02	3
Model B3b: (Model B1 + UMAP (max))	0.72 [0.65 - 0.81]	0.74	0.02	3
Model B3c: (Model B1 + UMAP (mean))	0.73 [0.65 - 0.81]	0.75	0.02	3
Model C1: (TIR and MAGE)	0.69 [0.61 - 0.77]	0.69	0.01	11
Model D1a: (UMAP, max*)	0.55 [0.48 - 0.64]	0.58	0.03	2
Model D1b: (UMAP, min*)	0.53 [0.43 - 0.63]	0.57	0.04	2
Model D1c: (UMAP, mean*)	0.53 [0.43 - 0.64]	0.56	0.03	2

* Indicates the pooling technique for the embeddings. * Abbreviations: * CI: Confidence Interval, CGM: Continuous Glucose Monitoring, UMAP: Uniform Manifold Approximation and Projection.

4.9 Study III: Results from an Algorithmic Fairness Study

Bias is an important spect to consider when working data drivenly. In study III, we therefore evaluates bias in CGM trained machine learning models for forecasting in the T1DX cohort

4.9.1 Bias in Machine Learning Models

The RMSE of the models are shown in Table 4.15 and Table 4.16. Base-Generalized and Transfer Learned models achieved lower prediction errors (RMSE: 2.00 to 2.06 mmol/L) than the naive last observation carried forward (LOCF)(RMSE: 2.41 and 2.45 mmol/L) and Base-Individual models (RMSE: 2.54 to 2.63 mmol/L) for both White and Black participants (Table 4.15 and Table 4.16).

Table 4.15: Root mean squared errors with 95% CIs for the different models evaluated on White participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
10			2.04 [1.95, 2.13]	2.02 [1.94, 2.11]
20			2.04 [1.95, 2.12]	2.02 [1.94, 2.10]
30			2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
40	2.45 [2.32, 2.57]	2.54 [2.38, 2.70]	2.03 [1.94, 2.12]	2.02 [1.94, 2.10]
50	(independent	(independent	2.03 [1.94, 2.11]	2.02 [1.93, 2.10]
60	of ratio)	of ratio)	2.02 [1.94, 2.11]	2.02 [1.93, 2.10]
70			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
80			2.02 [1.93, 2.11]	2.01 [1.93, 2.10]
90			2.01 [1.92, 2.10]	2.01 [1.93, 2.10]
100			2.01 [1.92, 2.10]	2.01 [1.92, 2.10]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40% children) and sex 57%*women* distribution.

Abbreviations: LOCF: last observation carried forward.

Table 4.16: Root mean squared errors with 95% CIs for the different models evaluated on Black participants.

Ratio	LOCF	Base-Individual	Base-Generalized	Transfer Learned
0			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
10			2.05 [1.96, 2.14]	2.00 [1.91, 2.08]
20			2.05 [1.96, 2.14]	2.00 [1.92, 2.08]
30			2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
40	2.41 [2.28, 2.53]	2.63 [2.47, 2.79]	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
50	(independent	(independent	2.05 [1.97, 2.14]	2.00 [1.92, 2.08]
60	of ratio)	of ratio)	2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
70			2.06 [1.97, 2.14]	2.00 [1.92, 2.08]
80			2.06 [1.97, 2.14]	2.00 [1.92, 2.09]
90			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]
100			2.06 [1.97, 2.15]	2.00 [1.92, 2.09]

Values that vary by proportion are estimated using linear mixed-effects models. These models were adjusted to the population's age (40

For the Base-Generalized models, prediction performance gradually diverged between White and Black participants as the proportion of White participants in the training data increased (Figure 4.4). For example, when the training data changed from 0% to 100% White participants, the difference in RMSE between the racial groups increased by 0.04 mmol/L [95% CI: 0.001 to 0.008], $p=0.02$, indicating that training data composition affected the two groups differently (Table 4.17). This divergence was not observed after transfer learning (0.002 mmol/L [95% CI: -0.001 to 0.005], $p=0.20$) (Table 4.18). A statistically significant differences were however found in both the Base-Generalized and the Transfer Learned models for age groups (0.482 [0.355 to 0.610], $p < 0.001$, Table 4.17 and 0.411 [0.290 to 0.533], $p < 0.001$, Table 4.18 respectively). The differences in age groups showed a lower error when forecasting blood glucose values for adults compared to children. Larger sample sizes available for transfer learning were associated with lower prediction error ($\beta = -0.111$ mmol/L [-0.161 to -0.061] mmol/L, $p < 0.001$) for every additional 1,000 samples used for fine-tuning (Table 4.18). In contrast, the size of the dataset used to train the Base-Generalized model was not associated with prediction error.

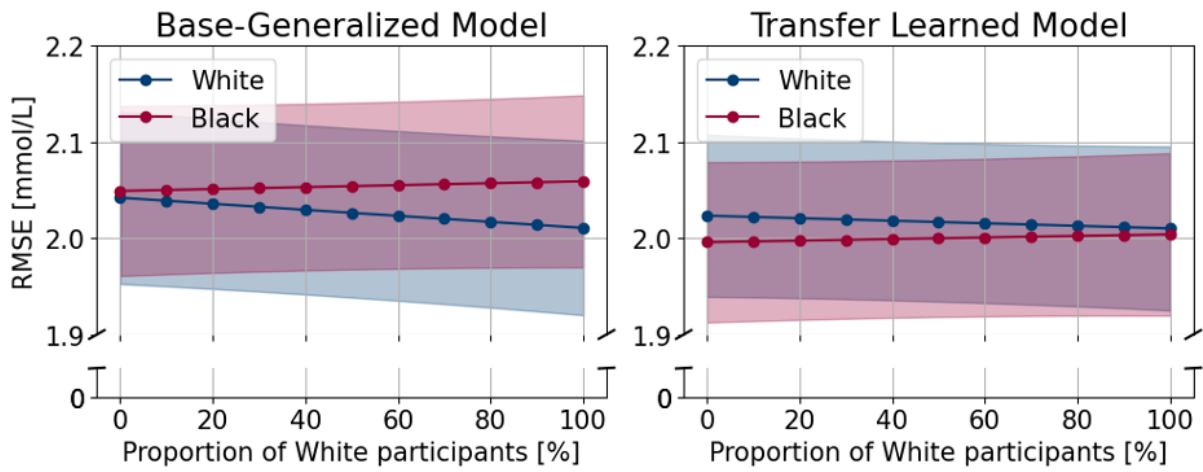


Figure 4.4: Estimated root mean squared errors (RMSE) across 11 different proportions of White and Black participants in the dataset.

Table 4.17: Model coefficients from linear mixed effects models for root mean squared errors of the Base-Generalized models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.893	[1.762, 2.024]	<0.001
Race			
White	Reference		
Black	0.007	[-0.118, 0.132]	0.91
Age			
Adult	Reference		
Child	0.482	[0.355, 0.610]	<0.001
Sex			
Male	Reference		
Female	-0.076	[-0.201, 0.048]	0.23
Training size Base-Generalized*	0.019	[-0.055, 0.093]	0.62
Ratio (per 10%)	-0.003	[-0.007, 0.001]	0.12
Ratio · Race	0.004	[0.001, 0.008]	0.02

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

Table 4.18: Model coefficients from linear mixed effects models for root mean squared errors of the Transfer Learned models.

Predictor	Coefficient	95% CI	P-value
Intercept	1.890	[1.768, 2.015]	<0.001
Race			
White	Reference		
Black	-0.028	[-0.147, 0.092]	0.65
Age			
Adult	Reference		
Child	0.411	[0.290, 0.533]	<0.001
Sex			
Male	Reference		
Female	-0.057	[-0.173, 0.060]	0.34
Training size Base-Generalized*	0.011	[-0.060, 0.081]	0.77
Training size tl**	-0.111	[-0.161, -0.061]	<0.001
Ratio (per 10 units)	-0.001	[-0.003, 0.001]	0.46
Ratio · Race	0.002	[-0.001, 0.005]	0.20

* Per 100,000 samples. (Training size was centered at 510,000 as the training data had a mean of 514,293 samples)

** Per 1,000 samples. (Training size for the transfer learned approach was centered at 4,400 as the training data for fine-tuning had a mean of 4,371 samples.)

4.9.2 The Surveillance Error Grid

Surveillance Error Grid analysis showed minimal differences between models and training data compositions. Across all models, racial groups, and training data ratios, almost all predictions fell within the “None” or “Slight” risk zones, while approximately 3–6% were classified as “Moderate” risk and no predictions were classified as “High” or “Extreme” risk (Appendix F). These findings indicate that the small differences in RMSE observed between models were unlikely to be clinically meaningful.

5 Discussion

Main findings 6

- Scoping review highlighted a chaotic landscape of wildly different study designs in the literature
- Mention other reviews, Yapanis + the other one that I forgot the name of
- Representation learning of CGM data can capture complex patterns in glucose dynamics that are not captured by traditional CGM metrics
- Machine learning models can predict NT-proBNP levels using CGM data, however the performance of these models is modest and there is still room for improvement
- Bias is a concern in machine learning models, and fine-tuning on a subset of the data can help mitigate bias as measured by fairness metrics
- Grouped fairness is easier to evaluate and mitigate than individual fairness, but it is still an important step in ensuring that machine learning models are equitable and do not perpetuate existing biases in healthcare.
- CGM contains clinically relevant information beyond glucose monitoring.
- Conventional CGM-derived metrics have been extensively investigated but evidence remains heterogeneous.
- Foundation models capture meaningful glycemic structure across individuals.
- Embeddings did not improve cardiovascular prediction in the available dataset.
- Fair machine learning requires attention to data composition, but transfer learning can mitigate disparities.
- Does CGM contain clinically relevant cardiovascular information?
- Study 1
 - Heterogeneous literature, mostly association studies and almost no prediction studies.
 - TIR and MAGE dominate the literature (Hopefully here add that many different outcomes for each CGM-derived metric)
- Study 2
 - TIR and MAGE predicted NT-proBNP
 - Why not use CONGA; AI: In participants with insufficient valid CGM observations, CONGA could not be calculated reliably using the available implementation
 - Embeddings captured glycemic control but no obvious cardiovascular structure
 - CVD is multifactorial and CGM captures only one component
 - cross-sectional data
- What can foundation models contribute with beyond conventional CGM metrics?
 - Capture more details patterns?
 - external generalizability to known publicly available model
 - reusable glucose representations and open access in the future
 - meaningful representations better prediction

- Machine learning in clinical prediction: complexity versus utility
 - Traditional metrics remain competitive.
 - Simpler models often performed as well.
 - Foundation models UMAPs might not be the best solution -> enough data to train a head?.
 - Robust validation is essential.
 - Optimism correction.
 - Small datasets.
- Fairness and generalizability of CGM-based machine learning
 - Study III belongs.
 - Why fairness matters
 - Small differences solved with TL
 - Surveillance error grid and clinical significance or lack thereof when focused in this model
 - Data diversity
 - Generalizability
 - Beyond race (age, sex, ethnicity, healthcare systems)
 - Many studies in the literature tests a variety of different prediction horizons but it seems to always predict best, the shorter the prediction horizon is. Most often sensors do also have a higher resolution then 15 minuts (3-5 minutes resolution) (I have no idea where to place this bit so placed with study 3 for now)
 - we attempt to mitigate bias by fine-tuning the model on a subset of the data. The results of this analysis are presented in the paper, and we find that fine-tuning the model can reduce bias as measured by fairness metrics. However, it is important to note that bias mitigation is an ongoing process, and further research is needed to develop more effective strategies for addressing bias in machine learning models in healthcare
- Data driven prediction models are only as useful as the populations on which they are developed and evaluated. (Methodological considerations)
 - Cross-sectional versus longitudinal studies
 - Something about data and cross-sectional vs longitudinal data and how this affects the choice of methods
 - Sample size
 - External validation
 - Choice of cardiovascular outcome
 - Heterogeneity
 - Open science
 - T1DX dataset (half kids, half adults have stronger signal because kids have poor glycemic control) and
 - CANCAN dataset (very specific T2D population)
 - Mention the CANCAN dataset and the weird trend with people being better of the more comobidities
 - Foundation model availability

Clinical implications and Future perspectives 7

- Current clinical utility of CGM-derived metrics
 - Where foundation models may become useful (Fine-tuning foundation models, Multi-modal foundation models, Raw CGM instead of summary metrics)
 - Importance of interpretable predictors
 - Implications for cardiovascular risk assessment
 - Personalized diabetes care
 - Fair AI and External validation -> Open foundation models
 - Larger prospective cohorts
 - mention protocol cited at the end of scoping review discussion

8 Conclusions

Conclusion 9

- CGM data has the potential to provide valuable insights into cardiovascular risk in diabetes, but there are still many challenges to overcome in terms of study design, data quality, and machine learning methods.

10 Perspectives

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Study I: Search strings A

Table A.1: Search in Ovid MEDLINE (Search date: 22 January 2024)

#	Searches	Results
1	Blood Glucose	10241
2	Self-Monitoring/ Continuous Glucose Monitoring/	34
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> cgmorb ^{gmoriscgmorr} tcgmor(<i>timeadj3range</i>)). <i>ti, ab, kf, kw</i> or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>)adj5(<i>flashorfluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 8669 5 <i>or/1—</i> 4 38366 6 <i>expCardiovascularDiseases/</i> 2760764 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 2182473 8 <i>or/6—</i> 7 3971342 9 <i>expDiabetesMellitus/</i> 518864 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i> or/9-10	815092
11	or/9-10	875527
12	and/5,8,11	2416
13	exp Animals/ not Humans/	5189493
14	12 not 13 Updated search: 11 March 2025	2300
15	limit 14 to (da="20231201-20261231" or dt="20231201-20261231" or ez="20231201-20261231" or ed="20231201-20261231")	310
16	or/14-15	2610

Table A.2: Search in Embase (Ovid) (Search date: 22 January 2024)

#	Searches	Results
1	blood glucose monitoring/	37574
2	exp continuous glucose monitoring system/	5648
3	((blood-glucose or bloodglucose or glucose) adj2 (monitor\$ or selfmonitor)) <i>or</i> <i>cgmorb</i> <i>gmoris</i> <i>cgmorr</i> <i>tcgmor</i> (<i>timeadj3range</i>)). <i>ti, ab, kf, kw.</i> 4850 or bloodsugar\$ or glucose\$ or glyc?emic\$ or sugar). <i>ti, ab, kf, kw.</i>) <i>adj5</i> (<i>flashorfluctuat</i> or intermittent\$ or scan\$ or variabilit). <i>ti, ab, kf, kw.</i> 16430 5 <i>or</i> /1–4 81510 6 <i>expcardiovascular</i> disease/ 5122163 7 (cardiovasc or cvd or cardiometabol\$ or cardio-metabol\$ or heart\$ or macrovasc\$ or vasc). <i>ti, ab, kw, kf.</i> 3044925 8 <i>or</i> /6–7 6303633 9 <i>expdiabetesmellitus</i> / 1239184 10 (diabet or dm1 or dm2 or dmi or dmii or iddm or niddm or t1d or t2d). <i>ti,ab,kw,kf.</i>	1237019
11	or/9-10	1501812
12	and/5,8,11	11217
13	(rat or rats or mouse or mice or swine or porcine or murine or sheep or lambs or pigs or piglets or rabbit or rabbits or cat or cats or dog or dogs or cattle or bovine or monkey or monkeys or trout or marmoset\$1). <i>ti.</i> and animal experiment/	1237966
14	animal experiment/ not (human experiment/ or human/)	2601109
15	or/13-14	2672240
16	12 not 15	10861
17	limit 16 to exclude medline journals Updated search: 11 March 2025	2176
18	limit 17 to (dc=“20231201-20261231” or dd=“20231201-20261231” or rd=“20231201-20261231”)	467
19	or/17-18	2643

Study I: Included and excluded reports **B**

C Study I: Geographical location

Asia (n=27) • China (n=19) • Japan (n=8) Europe (n=22) • Italy (n=7) • 3 Belgium (n=3) • 2 Ukraine (n=2) • 2 Finland (n=2) • 2 Spain (n=2) • 1 Greece (n=1) • 1 Malta (n=1) • 1 Sweden (n=1) • 1 The Netherlands (n=1) • 1 Portugal (n=1) • 1 Russia (n=1) North America (n=2) • 2 USA (n=2) Australia / Oceania (n=1) • 1 Australia (n=1)

Borg 2011(ADAG study) had study centers in: • Europe: Netherlands, Denmark, Italy • Africa: 2011 • North America: USA

Borg R, Kuenen J C, Carstensen B, Zheng H, Nathan D M, Heine R J, et al. HbA1(c) and mean blood glucose show stronger associations with cardiovascular disease risk factors than do postprandial glycaemia or glucose variability in persons with diabetes: the A1C-Derived Average Glucose (ADAG) study. 2011; ADAG Study Group, editor. Diabetologia. 54: 69–72. doi:10.1007/s00125-010-1918-2

Study I: Subclinical results

D

- ☐ Add table with subclinical study design overviews
- ☐ Add table overview of subclinical results

E Study II: Prediction stability

Prediction stability plots (?@fig-pred__stability) revealed more stability the fewer predictors.

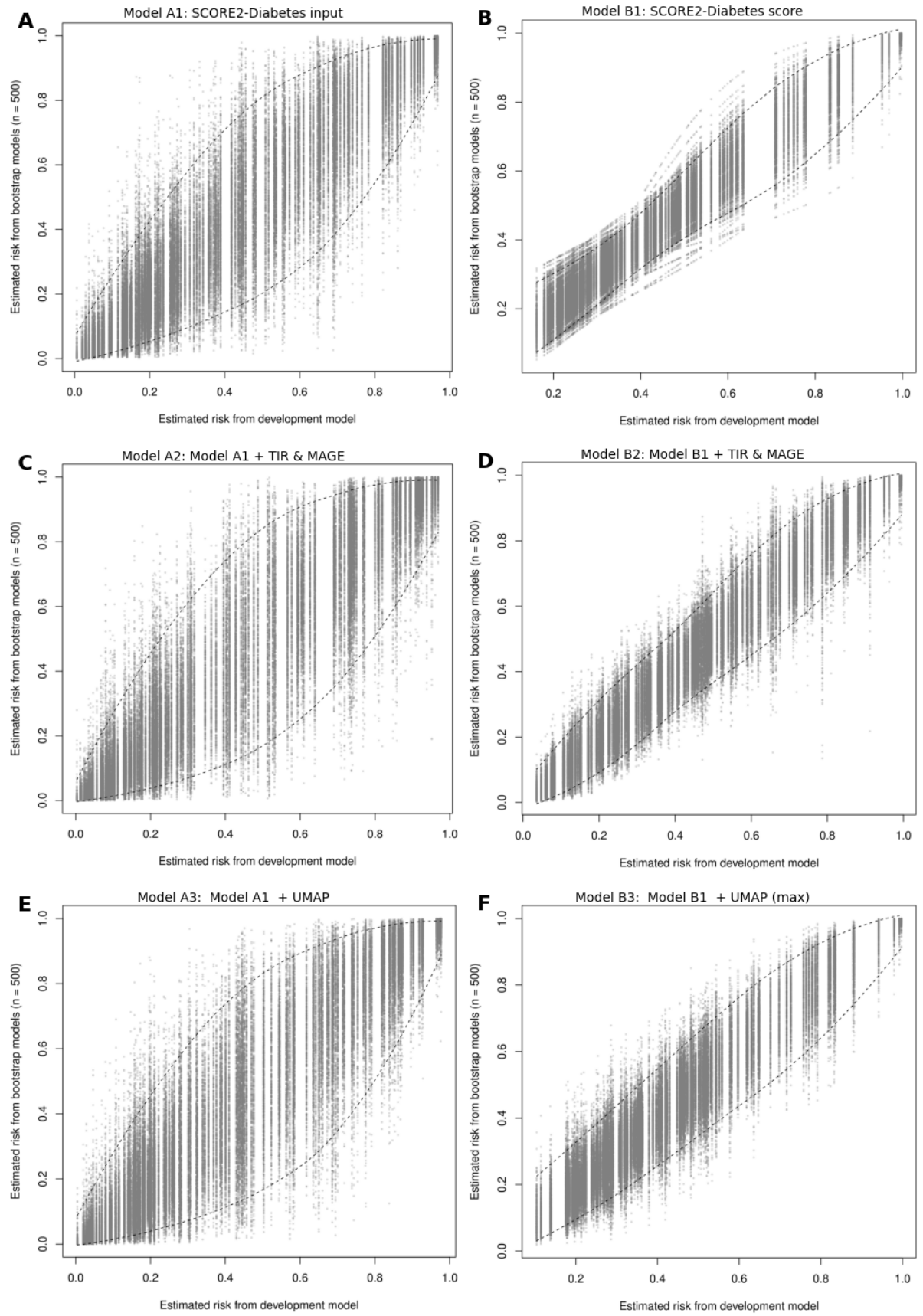


Figure E.1: Prediction stability plots show the predictions (estimated risk) for the full dataset from the development model (equivalent to the apparent model) on the x-axis, and the predictions from all bootstrap models on the y-axis. Tight alignment along the diagonal indicates higher prediction stability.

F Study III: Surveillance Error Grid Results

Table F.1: Results from the Surveillance Error Grid for White participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
White	LOCF	None	69%	69%	69%	69%	69%	69%
		Slight	26%	26%	26%	26%	26%	26%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	28%	28%	28%	28%	28%	28%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Generalized	None	72%	73%	73%	73%	73%	73%
		Slight	25%	25%	25%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Transfer Learned	None	72%	72%	73%	73%	73%	73%
		Slight	25%	24%	24%	24%	24%	24%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

□ maybe add this one to appendix, since I think they show the same things

Table F.2: Results from the Surveillance Error Grid for Black participants. Values are reported as percentages.

Group	Model	Zone	0	20	40	60	80	100
Black	LOCF	None	71%	71%	71%	71%	71%	71%
		Slight	24%	24%	24%	24%	24%	24%
		Moderate	4%	4%	4%	4%	4%	4%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%
	Base-Individual	None	66%	66%	66%	66%	66%	66%
		Slight	27%	27%	27%	27%	27%	27%
		Moderate	6%	6%	6%	6%	6%	6%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

Group	Model	Zone	0	20	40	60	80	100
	Base -Generalized	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	23%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
	Transfer Learned	Extreme	0%	0%	0%	0%	0%	0%
		None	75%	74%	75%	75%	74%	75%
		Slight	22%	23%	22%	22%	22%	22%
		Moderate	3%	3%	3%	3%	3%	3%
		High	0%	0%	0%	0%	0%	0%
		Extreme	0%	0%	0%	0%	0%	0%

G Full papers

G.1 Paper I

G.2 Paper II

G.3 Paper III

Addenda and errata



This chapter contains an overview of:

- Addenda: contents added since the submitted version of the dissertation.
- Errata: corrections to errors discovered since submission of the dissertation.

H.1 Addenda

H.2 Errata